

10.2 Malignant hyperthermia — full protocol

Item	Detail
Genetics	Autosomal dominant; RYR1 (ryanodine receptor) gene on chromosome 19 ; uncontrolled Ca ²⁺ release → sustained contracture and a hypermetabolic crisis
Triggers	All volatile inhalational agents + SUCCINYLCHOLINE. Safe: propofol, thiopentone, benzodiazepines, opioids, nitrous oxide, non-depolarising relaxants, all local anaesthetics
Earliest sign	Unexplained rise in EtCO₂ with tachycardia/tachypnoea; masseter spasm after succinylcholine. Hyperthermia is a LATE sign — do not wait for it
Full picture	Rigidity, hypercarbia, mixed acidosis, hyperkalaemia , arrhythmias, temperature rising 1-2 °C every 5 min, myoglobinuria/rhabdomyolysis , raised CK, DIC, renal failure
Manageme nt	1 Call for help and the MH cart • 2 stop all triggers , change circuit, hyperventilate with 100 % O₂ at ≥ 10 L/min • 3 DANTROLENE 2.5 mg/kg IV bolus, repeat every 5-10 min up to 10 mg/kg (may need 30) • 4 finish/abandon surgery quickly • 5 active cooling — iced IV saline, ice packs, cold lavage of cavities; stop at 38 °C • 6 treat hyperkalaemia and arrhythmias (avoid calcium-channel blockers with dantrolene) • 7 maintain urine output > 1-2 mL/kg/h • 8 monitor core temperature, ABG, K ⁺ , CK, coagulation • 9 ICU ≥ 24-36 h (recrudescence up to 25 %) • 10 counsel family, refer for caffeine-halothane contracture or genetic testing , issue an alert card
Dantrolene	20 mg lyophilised vials reconstituted with 60 mL sterile water ; keep a ~36-vial stock
Mortality	> 70-80 % untreated → < 5-10 % with prompt dantrolene

10.3 Anaphylaxis in the OR

- Commonest triggers: **neuromuscular blocking agents, antibiotics, chlorhexidine, LATEX, colloids, contrast.**
- **Latex anaphylaxis is typically DELAYED 20-60 min** after exposure (absorption across mucosa/peritoneum) — a favourite MCQ. Highest risk: **spina bifida, children with multiple operations, health-care workers. Cross-reacting foods: banana, avocado, chestnut, kiwi.**
- Signs under anaesthesia: **unexplained hypotension and tachycardia (commonest), high airway pressure / bronchospasm, desaturation**, flushing, angio-oedema, arrest.
- **Management: stop the trigger → 100 % O₂ and secure the airway → ADRENALINE 0.5 mg IM (0.5 mL of 1:1000) into the antero-lateral thigh, repeat every 5 min** (or **50 µg IV boluses** if monitored and experienced) → **rapid IV crystalloid 20 mL/kg, legs up → chlorpheniramine 10 mg, hydrocortisone 200 mg, nebulised salbutamol.** Take **mast-cell tryptase** samples and refer for allergy testing.
- **Latex-safe environment:** first case of the day, latex-free gloves/tubing/catheters, a dedicated **latex-free crash cart**, signage on the door.

10.4 Fire in the operating room

Element	Detail
Fire triangle	FUEL + OXIDISER + IGNITION (heat) — all three abound in an OR
Commonest ignition source	Electrosurgical unit (≈ 70-90 % of OR fires) , then laser, then fibre-optic light cables
Commonest fuel	Alcohol-based skin prep (pooled or not dry), then drapes, gowns, gauze, hair, ET tubes

SUPPLEMENTARY NOTES — EXAM-FOCUSED

Preparation of Patient for Operation • Pre & Post Operative Patient Care

Only the gaps in the studied PDF (Ch. 1 Perioperative Care of the Patient • Ch. 2 Protection of the Patient in Surgery • Ch. 3 Safety Measures for OR Personnel), arranged in syllabus order

SCOPE RULE USED FOR THIS DOCUMENT
Sections are numbered to match your syllabus sub-points exactly, so you can tick them off one by one.
Kept: every gap that sits under a listed sub-point — admission procedure, transfer/position, environmental controls, electrosurgery, operative records, counting, **sterilization**, emergencies & disasters, in-service education, body mechanics, fatigue, radiation safety, infection control, chemical hazards, plus **post-operative care** (named in the unit heading).
Cut as off-syllabus: ASA/Mallampati and cardiac-risk scoring, the routine-investigation list, premedication pharmacology, peri-operative anticoagulant/herbal management, ERAS, post-operative fluid and nutrition regimens, analgesic pharmacology in depth, discharge medication reconciliation, laser classes, local-anaesthetic toxicity, and the single-use-device reuse debate. These belong to anaesthesia, clinical-pharmacy or equipment units, not to this one.
Revise from the PDF, not from here: psychological support (Ch. 1 is excellent), the record forms, the position figures, the 9 counting rules, orientation and fatigue text. This document only adds what those pages leave out.

1. PRE-OPERATIVE CONSIDERATIONS — PSYCHOLOGICAL SUPPORT

Your PDF covers this sub-point thoroughly. Only four things are missing, and all four are one-line answers.

Point	Content
Sources of pre-operative anxiety	Fear of the unknown , of death , of not waking from anaesthesia , of pain , of disfigurement/altered body image , of loss of control and dependence ; plus separation from family and financial worry
Assessment tools	STAI (State-Trait Anxiety Inventory) • APAIS (Amsterdam Preoperative Anxiety and Information Scale) • Visual Analogue Anxiety Scale • m-YPAS for children
Peak anxiety	Immediately before transfer to the OR (also on being told surgery is needed, and the night before)
Effects of unrelieved anxiety	↑ catecholamines → tachycardia, hypertension, arrhythmia • ↑ anaesthetic requirement • ↑ post-operative pain scores and analgesic need • ↑ PONV • delayed wound healing • longer stay
Most effective intervention	Structured pre-operative teaching/counselling. Then relaxation and guided imagery, music, parental presence at induction in children, allowing spectacles/hearing aids/dentures to stay until induction
Age-specific fears	1-3 y separation • 3-6 y mutilation • school age loss of control • elderly dependency and death (needs slower, repeated explanation)

2. PREPARATION OF THE PATIENT FOR OPERATION (PHYSICAL PREPARATION)

The PDF reproduces the pre-operative checklist form but never explains the preparation itself.

2.1 Pre-operative fasting — remember 2 - 4 - 6 - 8

Ingested material	Minimum fast
Clear liquids (water, pulp-free juice, black tea/coffee)	2 hours
Breast milk	4 hours
Infant formula, non-human milk, light meal	6 hours
Fried/fatty food, meat, full meal	8 hours

WHY — and when the rule does not apply
Purpose = prevent **pulmonary aspiration of gastric contents (Mendelson's syndrome)** at induction, when airway reflexes are abolished. Risk rises when gastric volume is **> 25 mL (0.4 mL/kg)** and **pH < 2.5**.
Treat as a full stomach regardless of fasting time: emergency and trauma, pregnancy, obesity, diabetic gastroparesis, bowel obstruction, severe GERD, opioids → **rapid sequence induction with cricoid pressure (Sellick's manoeuvre)** and acid prophylaxis.
Prolonged ‘midnight’ fasting is harmful (dehydration, hypoglycaemia, insulin resistance) and does **not** reduce gastric volume.

2.2 Skin preparation and hair removal — evidence the PDF predates

Question	Answer
Should hair be removed?	No — not unless it physically interferes with the procedure or closure
If it must be removed?	Electric clippers with a single-use head , or depilatory cream. Razors are contraindicated (micro-abrasions become colonised)
When?	Immediately before surgery, outside the OR . SSI risk: no removal ≈ clipping < shaving immediately before < shaving the night before (highest)
Best antiseptic	Alcohol-based 2 % chlorhexidine gluconate > aqueous povidone-iodine. Povidone-iodine 10 % = 1 % available iodine and needs 2-3 min contact and must dry
Alcohol + ESU	Prep must dry completely (≥ 3 min) and must never pool — pooled alcohol is the classic fuel in OR fires (§10)
Direction	From the incision outward in concentric strokes; the sponge is never returned to the centre ; the dirtiest area (stoma, sinus, perineum, umbilicus, axilla) is prepped last
Extent	Wide enough to allow extension of the incision, new incisions and drain sites without re-prepping
Contraindications	Chlorhexidine — middle ear (ototoxic), eye (keratitis), neural tissue ; iodophors — neonates, iodine allergy

2.3 Immediate pre-transfer preparation

- Patient **voids** immediately before transfer; catheterise only if indicated, and **remove within 24 h** (CAUTI prevention).
- **Remove and document the disposition of:** dentures/bridges, contact lenses, jewellery (tape a wedding ring if it cannot be removed), **nail polish and artificial nails** (interfere with pulse oximetry and harbour organisms), hairpins, prostheses, valuables — exactly what Form 1-1 records.
- **Antibiotic prophylaxis: single IV dose within 60 minutes of skin incision** (120 min for vancomycin); re-dose if surgery exceeds 2 half-lives or blood loss > 1500 mL; **stop within 24 h**. For tourniquet surgery the infusion must finish **before inflation**.

Type	What it is	What it proves	Examples
Chemical (ISO 11140, classes 1-6)	Heat/chemical-sensitive dyes on tape, strips, cards	Exposure to the process — NOT sterility	Class 1 process indicator (external autoclave tape) • Class 2 = Bowie-Dick • Class 3 single-variable • Class 4 multi-variable • Class 5 integrating (all critical variables, correlates with the BI) • Class 6 emulating (most stringent). An internal indicator goes in the centre of every pack
Biological (BI) — GOLD STANDARD	Calibrated highly resistant bacterial spores processed with the load, then incubated	The only indicator that proves the process actually killed spores	Steam & H₂O₂ plasma → Geobacillus stearothermophilus (incubate 55–60 °C) • Dry heat & EO → Bacillus atrophaeus (35–37 °C) • γ-radiation → Bacillus pumilus

PACKAGING, STORAGE AND THE PROCESSING SEQUENCE
Packaging must **allow sterilant penetration, be a microbial barrier, permit aseptic opening and resist moisture**: double-thickness muslin, non-woven polypropylene, crepe paper, paper-plastic peel pouches, rigid containers with filters. **Never closed metal containers or foil for steam**.
Shelf-life is EVENT-RELATED, not TIME-RELATED — a pack is sterile **until an event compromises it** (wet, torn, punctured, soiled, seal broken, dropped, over-handled). The old “sterile for 30 days” rule is obsolete.
Storage: clean, dry, dust-free, closed cupboard, **20-23 °C, humidity 30-60 %, ≥ 20-25 cm above the floor, 45 cm below the ceiling/sprinkler, 5 cm from outside walls**; stock rotated **FIFO**.
Processing sequence: point-of-use enzymatic spray in the OR → closed transport → **decontamination/washer-disinfector → ultrasonic cleaner (cavitation reaches box locks, serrations and lumens)** → rinse and dry → inspect, lubricate, assemble with **hinges and box locks OPEN** → pack with an internal indicator → sterilize → cool/dry → store.

10. EMERGENCIES AND DISASTERS

10.1 Cardiac arrest in the OR — the numbers

Item	Value
Sequence	C-A-B ; high-quality compressions are the priority
Rate / depth	100-120 per minute; ≥ 5 cm and ≤ 6 cm in adults (⅓ of chest depth in children)
Compression : ventilation	30 : 2 unintubated adult; continuous compressions with 1 breath every 6 s once an advanced airway is in; 15 : 2 two-rescuer paediatric
Quality	Interruptions < 10 s, full recoil , change compressor every 2 min , compression fraction > 60-80 % ; EtCO₂ < 10 mmHg = poor CPR, a sudden rise suggests ROSC
Defibrillation	Biphasic 120-200 J (monophasic 360 J). Shockable = VF and pulseless VT only; PEA and asystole are NOT shockable
Drugs	Adrenaline 1 mg IV every 3-5 min; amiodarone 300 mg then 150 mg for refractory VF/pVT
Reversible causes	4 H's — Hypoxia, Hypovolaemia, Hypo/Hyperkalaemia-metabolic, Hypothermia; 4 T's — Tension pneumothorax, Tamponade, Toxins, Thrombosis
Crash cart	Defibrillator + monitor , airway drawer (laryngoscope, ET tubes, LMA, oral airways, suction, bag-valve-mask), in-date emergency drugs , IV access and fluids, checked and re-sealed after every use and each shift with a signed checklist
Paediatric	PALS competency required; shock 2 J/kg then 4 J/kg ; adrenaline 0.01 mg/kg
Documentation	Every drug, dose, route, time, intervention and the person giving it — undocumented care is legally non-existent

- **45-50 °C, cycle ≈ 1 hour; by-products are only water vapour and oxygen → NO aeration or cooling needed**, no toxic residue.
- Ideal for **fibre-optic cables, endoscopes, microsurgical and electrical equipment, glass, ceramics, metals**.
- **Cannot be used for cellulose — paper, cotton, linen, gauze, dressings** (they absorb the sterilant) — nor for liquids, powders or long narrow lumens. High capital cost.
- **Biological indicator: Geobacillus stearothermophilus.**

9.7 Liquid chemical sterilization / high-level disinfection

Agent	Sterilization	High-level disinfection	Notes
Glutaraldehyde 2 % (activated — Cidex)	10 hours	20-45 min at 20-25 °C (your PDF says 10 min — quote the textbook if the question is from it)	Does not corrode lenses or metal → endoscopes. Rinse in sterile distilled water. Test potency with a strip before each use. Irritant/sensitiser — asthma, dermatitis; ceiling 0.05 ppm; wear polyethylene or double gloves
OPA 0.55 %	—	12 min at 20 °C	Faster, less irritant; stains skin grey
Peracetic acid 0.2 % (Steris)	12 min at 50-56 °C	—	Single-use, automatic dilution, effective even with organic soil, low surface tension → reaches lumens; decomposes to acetic acid + water + oxygen
Hydrogen peroxide 7.5 %	6 hours	30 min	Non-toxic breakdown products

9.8 Radiation and filtration

Method	Detail
Ionising (γ) radiation	Cobalt-60, dose 25 kGy (2.5 Mrad). A cold, industrial method for pre-packed single-use disposables — syringes, needles, gloves, catheters, sutures, blades, heart valves, bone grafts. Not used inside hospitals. BI = Bacillus pumilus
Non-ionising UV-C (254 nm)	Surface and air disinfection only — poor penetration (blocked by glass, paper, water film). A disinfectant, not a steriliser. Hazard: keratoconjunctivitis and erythema — never run with the room occupied
Filtration	0.22 μm membrane for heat-labile liquids (sera, antibiotic and vaccine solutions, media). Removes but does not kill, and does not retain viruses or mycoplasma (needs 0.1 μm). Air equivalent = HEPA (\$5.1)

9.9 Monitoring — the three-indicator system

Type	What it is	What it proves	Examples
Physical / mechanical	Gauges, printout, chart recorder, cycle log	That the machine reached the parameters; read and signed every cycle	Sterilizer printout

- **Site marked** by the operating surgeon with an indelible marker, with the awake patient participating (§3).
- Confirm **consent, allergies, fasting status, prostheses/implants, blood availability** and that imaging and old records are in the OR.

3. ADMISSION PROCEDURE & PATIENT IDENTIFICATION

The PDF teaches the Joint Commission Universal Protocol in full. The **WHO Surgical Safety Checklist** is absent, and it is the internationally examined instrument.

3.1 WHO Surgical Safety Checklist (2008) — 3 phases, 19 items

Phase	When	Core content
1. SIGN IN	Before induction of anaesthesia	Patient confirms identity, site, procedure, consent • site marked • anaesthesia safety check • pulse oximeter on and working • known allergy • difficult airway / aspiration risk • risk of blood loss > 500 mL (7 mL/kg in children) → access and fluids planned
2. TIME OUT	After induction, before skin incision	All members state name and role • surgeon, anaesthetist and nurse verbally confirm patient, site, procedure • anticipated critical events (critical steps, duration, blood loss; anaesthetic concerns; sterility indicator and equipment issues) • antibiotic given in the last 60 min? • imaging displayed • VTE prophylaxis • correct position
3. SIGN OUT	Before the patient leaves the OR	Name of the procedure recorded • instrument, sponge and needle counts correct • specimens labelled and read aloud • equipment problems • key concerns for recovery

DISTINGUISH THESE THREE — favourite MCQ
Universal Protocol (JC, July 2004) = pre-procedure **verification** + **site marking** + **‘time out’** immediately before starting. Aim: prevent wrong-site, wrong-procedure, wrong-person surgery.
AORN (March 2005) added confirmation of the **correct patient position** to the time out.
WHO Checklist (2008) = the **3-phase, 19-item** tool; broader — it also covers airway, blood loss, antibiotics, imaging, counts and specimen labelling. It cut mortality from 1.5 % to 0.8 %.
Site marking: by the **operating surgeon, indelible marker**, at/near the incision, **with the awake patient participating, before** entering the OR; mark **laterality and level**; never mark the non-operative side.
At least two identifiers, and the patient **states** their own name — never “Are you Mary Jones?”

4. TRANSFER PROCEDURE & POSITION

4.1 Safe transfer — the numbers

- Minimum **two persons; ≥ 4 or a mechanical lifter** for an anaesthetised, obese or unstable patient; **one person on each side** of the table; **lock the wheels** of gurney and table.
- Devices that reduce the load: **roller board (Davis), slide sheet, Hoyer pad, air-assisted transfer mattress**.
- **NIOSH maximum recommended lift = 51 lb (23 kg)**; for **patient handling the limit is 35 lb (16 kg)** — above that, use an aid or a lift team. **Back injury is the commonest occupational injury in health care**.
- **Safety strap across the mid-thighs, ≈ 3 inches (7.5 cm) above the knees** — the single most important measure against a fall; two fingers should slide under it.

- Protect during transfer: **airway/ET tube, IV, arterial and CVP lines, epidural, Foley, drains** — and the **fingers** when table sections are raised.
- **Never move an anaesthetised patient without the anaesthetist's permission and direction.** The anaesthetist guards head/airway/neck, the surgeon protects fractures, the circulator applies padding and the strap.

4.2 Position-specific complications — the nerve-injury map

The PDF describes each position well but omits the complications, which is what gets asked.

Position	Nerve / structure at risk	Prevention	Physiological effect
Supine	Ulnar nerve — commonest peri-operative nerve injury; brachial plexus; radial n.; occiput, sacrum, heels	Arms abducted $\leq 90^\circ$, palms up (supinated) , elbows padded, legs uncrossed	↓ FRC; aorto-caval compression in late pregnancy
Trendelenburg	Brachial plexus (shoulder braces), retinal/cerebral congestion, sliding	Braces over the acromio-clavicular area, never on the soft neck ; limit to 30–45° and to the shortest time	↑ venous return , ↑ ICP , ↑ IOP , ↑ CVP ; ↓ FRC and compliance → atelectasis ; facial/laryngeal oedema; tube migration; regurgitation
Reverse Trendelenburg / sitting	Sciatic stretch, foot drop, venous pooling; venous air embolism in the sitting position	Padded footboard, antiembolism stockings + compression device, slow position change	↓ venous return → hypotension ; ↓ cerebral perfusion
Lithotomy	Common peroneal nerve — commonest here → foot drop ; femoral, obturator, saphenous, sciatic; lower-limb compartment syndrome	Pad the fibular head away from the post; raise and lower both legs together, slowly, by two people ; avoid extreme abduction; limit time (risk rises beyond 2-4 h)	On lowering the legs → sudden ↓ venous return → hypotension ; ↓ FRC
Prone	Brachial plexus, ulnar; eye → corneal abrasion and post-operative visual loss (POVL) from ischaemic optic neuropathy ; breasts, genitalia, iliac crests	Chest rolls from clavicle to iliac crest ; head on a padded donut/horseshoe with the eyes free of all pressure ; neutral neck; re-check the ET tube after turning	↑ abdominal and epidural venous pressure → bleeding; ↓ cardiac index
Lateral (kidney / thoracotomy)	Dependent brachial plexus and axillary vessels, dependent common peroneal and ear	Axillary roll caudal to the axilla — never in it ; pillow between the legs; head in line with the spine	V/Q mismatch — dependent lung better perfused but less ventilated ; kidney rest ↓ venous return
Jackknife / Kraske (a PRONE modification)	Face, eyes, genitalia, knees	Padding at hips, pillow under pelvis and shins	Marked ↓ venous return and ↓ FRC
Fracture table	Perineal post → pudendal nerve injury , perineal pressure necrosis	Well-padded post, minimum effective traction, protect the well leg; no skin-metal contact (ESU burn)	Blood loss; fat-embolism risk

- **Bowie-Dick test** = a **daily test of air removal / steam penetration in a PREVACUUM sterilizer**, run in an **empty chamber at 134 °C for 3.5 min**. It is a **Class 2 chemical indicator** and it does **not** prove sterility.
- **Biological indicator at least weekly (daily preferred) and with EVERY implant load** — implant loads are quarantined until the result is available. Vacuum leak test weekly.
- **Cannot be autoclaved:** sharp cutting edges (blunted), oils, greases, powders, anhydrous materials, low-melting plastics, lensed endoscopes, heat-labile electronics.
- Drying is part of the cycle — **a wet pack is a contaminated pack**; do not touch or move packs until cool and dry.

9.4 Dry heat (hot air oven)

Mechanism: oxidation — needs much higher temperature and longer time than moist heat.

Temperature	Holding time
160 °C	2 hours (the pair most often asked)
170 °C	1 hour
180 °C	30 minutes

- **Uses:** glassware, **sharp instruments (does not blunt or corrode them)**, oils, greases, powders, anhydrous fats, glass syringes. **Not for** rubber, plastics or fabrics.
- Load loosely for air circulation; **allow to cool before opening**.
- **Other dry-heat methods:** **incineration** ($\geq 1100\text{--}1200\text{ °C}$, final disposal of contaminated and anatomical waste), **red-heat flaming** (loops), **glass-bead steriliser** (250 °C, 20–40 s).
- **Biological indicator:** **Bacillus atrophaeus**.

9.5 Ethylene oxide gas

Parameter	Value
Mechanism	Alkylation of DNA, RNA and protein
The 4 critical variables	Gas concentration 450–1200 mg/L • temperature 37–63 °C • relative humidity 40–80 % • exposure 1–6 h (the PDF quotes 3–7 h cycles) — all four must be met
Aeration is mandatory	Mechanical aerator 8–12 h at 50–60 °C (air changed ≥ 4 times/h), or a ventilated room for 7 days at 18–22 °C . Residual EO causes chemical burns and haemolysis
Uses	Heat- and moisture-sensitive items — flexible endoscopes, cardiac catheters, plastics, electrical and electronic equipment, prosthetic valves, pacemakers, delicate optics
Rule	Anything that can withstand steam should NOT be gas-sterilized
Hazards	Toxic, irritant, mutagenic, teratogenic, carcinogenic (leukaemia/lymphoma), flammable and explosive; OSHA PEL 1 ppm (8-h TWA) , excursion 5 ppm/15 min
Biological indicator	Bacillus atrophaeus
Note on the indicator	An EO chemical indicator shows the item was exposed to EO — it does not prove sterilization occurred

9.6 Low-temperature hydrogen peroxide gas plasma (STERRAD)

- **58 % hydrogen peroxide** under vacuum, then excited by **radiofrequency energy** into a plasma of free radicals → kills bacteria, fungi, viruses **and spores**.

Term	Definition
Asepsis	Medical asepsis = clean technique (reduce numbers); surgical asepsis = sterile technique (absence of all organisms)
Sporicidal	Kills spores — the property that makes an agent a sterilant rather than a disinfectant
SAL (Sterility Assurance Level)	10⁻⁶ — probability of ≤ one viable organism in 10 ⁶ items. The definition of ‘sterile’
D-value	Time or dose that reduces the population by 1 log₁₀ (90 %)

9.2 Spaulding classification

Category	Definition	Examples	Processing
Critical	Enter sterile tissue or the vascular system	Surgical instruments, implants, needles, cardiac and urinary catheters, laparoscopes	Sterilization
Semi-critical	Contact intact mucous membranes or non-intact skin	Flexible endoscopes, laryngoscope blades, ET tubes, respiratory/anaesthesia equipment	High-level disinfection
Non-critical	Contact intact skin only	BP cuffs, stethoscopes, OR table, floors, furniture	Low/intermediate-level disinfection

Resistance to killing, most → least: Prions > bacterial spores > coccidia > mycobacteria > small non-lipid viruses > fungi > vegetative bacteria > lipid-enveloped viruses (HIV, HBV — the EASIEST to kill). A common MCQ inverts the last point. Prions: resist routine autoclaving — need 1 N NaOH plus 134 °C for 18 min, and single-use instruments where possible.

9.3 Moist heat — saturated steam under pressure (autoclave)

Mechanism: denaturation and coagulation of proteins. Moist heat beats dry heat because latent heat of vaporisation is released when steam condenses. It is the method of choice — fastest, cheapest, most reliable, non-toxic.

Cycle	Temperature	Pressure	Hold time	Use
Gravity displacement	121 °C (250 °F)	15 psi	15-20 min (wrapped 30 min)	General load, glassware, rubber, media
Gravity — high temp	134 °C	30 psi	3-3.5 min	Rapid cycles
Prevacuum (high-vacuum)	132-135 °C	≈ 30 psi	3-4 min + drying	Wrapped trays, textiles, lumened items — air is actively evacuated
Flash / IUSS	132 °C (270 °F)	≈ 27-30 psi	3 min unwrapped metal; 10 min porous/lumened	Only an urgently needed unwrapped single item — never implants or whole sets
Prions (CJD)	134 °C	—	18 min	After NaOH immersion

- **Air is the enemy of the autoclave** — trapped air lowers the temperature at a given pressure. So: **do not overload, do not wrap tightly, use mesh-bottom trays, place packs on edge with space between them, and never position a bowl so that it traps air.**

POSITIONING — GOLDEN RULES

Position is determined by the **surgical approach + patient's condition + anaesthetic technique + surgeon's preference**. Positioning aids must be **in the room before the patient arrives**. **PADDING** — Pressure points padded, **A**lignment maintained, **D**ignity/exposure minimised, **D**eveloped ready beforehand, **I**V lines protected, **N**erves free of stretch and compression, **G**entle simultaneous movement. Modifications of **supine**: Trendelenburg, reverse Trendelenburg, Fowler's/sitting, lithotomy. Modification of **prone**: Kraske/jackknife. **Compartment syndrome after lithotomy** = pain out of proportion, tense swollen calf, paraesthesia → **fasciotomy**. Always examine and document limb movement and paraesthesia on arrival in PACU.

5. ENVIRONMENTAL CONTROLS

5.1 Operating room design — the numbers the PDF omits

Parameter	Standard	Exam point
Temperature	20-23 °C (68-75 °F)	Team comfort vs patient hypothermia ; raise for neonates and burns
Relative humidity	20-60 % (classically 30-60 %)	Too low → static spark risk; too high → microbial growth and condensation
Air changes per hour	Minimum 15 , recommended 20-25 (≥ 3 outdoor air)	Dilutes contaminants and waste anaesthetic gas
Air pressure	POSITIVE relative to corridors	Air flows out of the OR. Contrast: TB isolation = NEGATIVE pressure
Airflow	Unidirectional downward over the table, exhaust at floor level	Keeps a clean air column over the sterile field
HEPA filter	≥ 99.97 % of particles ≥ 0.3 µm	The single most asked number. ULPA = 99.999 % at 0.1 µm (smoke evacuators)
Laminar / ultraclean flow	300-600 air changes/h	Used for joint arthroplasty and implant surgery
Noise	< 45 dB recommended	Silence is mandatory during induction — hearing is the last sense to be lost

5.2 Zoning of the OT complex

Zone	Also called	Areas	Attire
Protective / outer	Unrestricted	Reception, changing rooms, offices, pre-op holding, PACU	Street clothes permitted
Clean	Semi-restricted	Peripheral corridors, storage, utility, instrument processing	Scrub attire + head cover ; mask not required
Sterile / aseptic	Restricted	Operating room, scrub area, sterile store	Scrub attire + head cover + mask ; sterile gown and gloves in the field
Disposal / dirty	Dirty corridor	Dirty utility, sluice, disposal corridor	Scrub attire + PPE

5.3 Cleaning — and why fumigation is obsolete

- **Between cases:** all horizontal surfaces, the table, positioning devices, lights, ESU, suction and the floor **within 1-1.5 m of the field** are wiped with a hospital germicide. The PDF's key point: **every case is cleaned identically** — special ‘ritual’ cleaning after ‘dirty’ cases is **obsolete**, because all cases are treated as potentially infectious.
- **Terminal (daily) clean:** whole room, walls, ceiling, equipment wheels, scrub sinks, corridors, storage; wet-vacuum the floor.
- **Formaldehyde fumigation is no longer recommended** — Group 1 carcinogen, mucosal irritant, needs the room sealed and idle 24–48 h, and penetrates crevices unreliably. Modern no-touch options: **vaporised hydrogen peroxide, chlorine dioxide, UV-C** — always **after** physical cleaning, never instead of it.
- **Routine environmental culturing is not recommended** — only for outbreak investigation or after construction.
- **Blood spill:** absorb, then disinfect with **1 % sodium hypochlorite (10 000 ppm)** for 10 min.

5.4 Levels of disinfection (needed for §9 as well)

Level	Kills	Agents	Use
Sterilization	All microbes including spores	Steam, EO, H ₂ O ₂ plasma, glutaraldehyde 10 h, peracetic acid, radiation	Critical items
High-level (HLD)	All except large numbers of spores	Glutaraldehyde 2 %, OPA 0.55 % (12 min) , H ₂ O ₂ 7.5 % (30 min), peracetic acid	Semi-critical — endoscopes, laryngoscope blades, respiratory equipment
Intermediate	Vegetative bacteria, M. tuberculosis , most viruses/fungi; not spores	Alcohols 60–90 %, hypochlorite, iodophors, phenolics	Soiled non-critical surfaces
Low	Most vegetative bacteria; not TB, not spores	Quaternary ammonium compounds	Non-critical — floors, walls, BP cuffs, furniture

Key agent facts: alcohol — fastest kill, no residual action, not sporicidal, inactivated by organic matter, flammable. Chlorhexidine — persistent residual activity, not inactivated by blood, but ototoxic and corneal-toxic. Povidone-iodine — broad but inactivated by blood/pus and needs contact time to release free iodine. Hypochlorite — broad including spores, but corrosive and never mixed with acids or ammonia (chlorine gas). Quaternary ammonium compounds — inactivated by soap, hard water and gauze.

5.5 Aseptic technique and the sterile field — hard rules

1. Only **sterile items** are used in a sterile field.
2. The gown is sterile **in front from chest to the level of the field**, and the sleeves from **5 cm above the elbow to the cuff**. The **back is never sterile**.
3. A **draped table is sterile only at table-top level** — anything below the edge is unsterile and is **never brought back up**.
4. Sterile persons keep hands **at or above waist level, in front of the body**, and pass **back-to-back** or **face-to-face**.
5. Unsterile persons stay **≥ 30 cm (1 ft)** away, never reach over the field, and always **face** it.

MEDICO-LEGAL DOCTRINES — one-liners
Negligence — the 4 D's: Duty, Dereliction (breach of the standard of a reasonably competent practitioner — the **Bolam test**), Direct causation, Damage.
Res ipsa loquitur ('the thing speaks for itself') — negligence presumed; classic examples are a **retained sponge or instrument**, wrong-site surgery, and a diathermy burn.
Vicarious liability / respondeat superior — the hospital is liable for its employees' acts.
Battery = operating with **no** consent. **Negligence** = operating with **inadequately informed** consent.

8. COUNTING PROCEDURE

Learn the 9 counting rules from the PDF. What it lacks is the terminology and epidemiology.

- A retained item is a **‘never event’ / sentinel event**. The **commonest retained item is a sponge/gauze (≈ 50-70 %)**, then instruments and needles; commonest site = **abdomen/pelvis**.
- A retained gauze encased in a foreign-body granuloma = **gossypiboma (textiloma)** — presents weeks to years later with a mass, pain, sinus, abscess, obstruction or fistula, and is a classic **res ipsa loquitur** claim.
- **Risk factors:** emergency surgery, **unexpected intra-operative change of procedure**, high BMI, multiple teams or staff changes, and omitted or incorrect counts.
- **Counts are taken:** (1) **before the procedure (baseline)**, (2) when **items are added**, (3) **before closure of a cavity within a cavity** (uterus, bladder, bowel), (4) when **wound/fascial closure begins**, (5) at **skin closure**, and (6) at **any change of scrub or circulating personnel**.
- **The circulator records the count — that record is a legal responsibility.** Both persons count **aloud together** as the scrub person touches each item.
- All sponges used in a wound must be **radio-opaque (X-ray detectable)**; **X-ray-detectable sponges are never used as dressings**, and dressing sponges are never used in the wound. Adjuncts: **bar-coded counting and radiofrequency (RF) tag detection wands**.
- **If the count is incorrect:** repeat it → **inform the surgeon**, who explores the wound → if still missing, **intra-operative X-ray before the patient leaves the OR** → notify the OR supervisor → **document and file an incident report**. **Nothing (linen, trash, instruments, sponges) leaves the room until the final count is correct.**

9. STERILIZATION — HIGHEST-YIELD SECTION IN THE UNIT

9.1 Definitions — get these exactly right

Term	Definition
Sterilization	Destruction or removal of all forms of microbial life, including bacterial and fungal spores . An absolute term
Disinfection	Destruction of pathogens but not necessarily spores ; used on inanimate objects
Antisepsis	Destruction/inhibition of organisms on living tissue
Decontamination	Making an item safe to handle — always the FIRST step , before disinfection or sterilization
Cleaning	Physical removal of soil — a dirty instrument cannot be sterilized (organic matter blocks the sterilant)

7.2 What makes the record legally sound

- **JC-mandated minimum content:** surgeon's name and assistants, **pre-operative diagnosis**, procedure performed, **post-operative diagnosis**, findings, **specimens removed (type and number)**, estimated blood loss, and the names of all team members giving intra-operative care.
- Also document: the **time out**, the **counts and their outcome**, **skin condition before and after prep** and the solution used, position and positioning aids, **ESU pad site**, **serial number and settings**, tourniquet data, drains/catheters/packing, medications and irrigations (name, strength, dose, route, time, who gave them), implants, intake/output, transfusion checks, and any **unusual event with the actions taken**.
- **Rules:** contemporaneous, legible, factual and objective (no opinion or blame), signed and timed; **correct by a single line-through with initials, date and time — never erase, overwrite or use correction fluid**; leave no blank spaces; label late entries.
- **Incident (occurrence) report** for: incorrect count, medication error, burn, fall, retained item, equipment failure, needle-stick, lost specimen, unplanned re-operation and **death in the OR**.
- **Only what is documented is a legal fact** — the most quoted line in your PDF (pp. 8 and 48). An undocumented event ‘did not happen’ in court.

7.3 Consent — the parts the forms do not explain

Point	Rule
Five elements of valid consent	Disclosure • Capacity • Voluntariness • Comprehension • Documentation
Must be disclosed	Diagnosis, nature and purpose of the procedure, risks and benefits, alternatives including doing nothing , and consequences of refusal. No guarantee of outcome may be given
Timing	Before sedative premedication — hence the wording on Form 1-2, “I am not under the influence of any preoperative or other sedating medication”
Age	18 years in India (IPC §§ 87-88). For a child under 12 or a person of unsound mind, the guardian consents (IPC § 89)
Emergency / unconscious	Doctrine of necessity — IPC § 92: life-saving treatment in good faith without consent when the patient cannot consent and no representative is available
Consent obtained by fear or misconception	Invalid — IPC § 90
Special consent needed for	Sterilization (patient must state understanding of permanent, irreversible loss of fertility — Form 1-3), blood products, photography, implants, organ donation, HIV testing
Who obtains vs who verifies	The operating surgeon obtains it. The perioperative practitioner verifies that a valid, signed, dated consent exists and matches the procedure, site and side, and reports any discrepancy
Withdrawal	May be withdrawn at any time while the patient is competent
Blood refusal	Honour it, document it on the checklist , and plan cell-saver/tranexamic acid

- 6. The **edge of any wrapper, package or basin is unsterile** — conventionally a **2.5 cm (1 inch)** margin.
- 7. **Moisture contaminates by capillary action (strike-through)** — a wet field is a contaminated field.
- 8. Packages are opened **farthest corner first, nearest corner last**, held away from the body, so you never reach over the contents.
- 9. Sterile items stay **in view**; anything of **doubtful sterility is unsterile** (‘when in doubt, throw it out’).

5.6 Surgical hand antisepsis

Item	Standard
Agents	4 % chlorhexidine gluconate or 7.5 % povidone-iodine , or an alcohol-based surgical hand rub (WHO now prefers the rub)
Duration	First scrub of the day 3-5 minutes ; older counted-brush-stroke method = 30 strokes to nails, 20 per surface
Sequence	Nails → fingers and interdigital spaces → hands → wrists → forearms to 5 cm above the elbow ; the cleanest part (hand) is held highest ; rinse fingertips to elbow; never return to a cleaned area
Before scrubbing	Remove rings, watches, bracelets; nails short and clean, no polish, gel or artificial nails (AORN — they harbour Gram-negatives and fungi); do not scrub with cuts, abrasions or dermatitis
After scrubbing	Hands above the elbows, in front of the body, above waist and below shoulder level ; dry with a sterile towel from fingers to elbow, never back
WHO 5 Moments	1 before touching a patient • 2 before a clean/aseptic procedure • 3 after body-fluid exposure risk • 4 after touching a patient • 5 after touching patient surroundings
Rub vs soap	Alcohol rub routinely; soap and water is mandatory for visibly soiled hands and for Clostridioides difficile (alcohol is not sporicidal)
Double gloving	Reduces inner-glove perforation by $\approx$ 70-80 % ; change every 90-150 min and between patients

5.7 Surgical site infection — classification

Wound class	Definition	Example	SSI risk
I - Clean	No inflammation; respiratory/GI/GU tract not entered ; closed primarily	Thyroidectomy, hernia repair, joint replacement	< 2 %
II - Clean-contaminated	Tract entered under controlled conditions , no unusual contamination	Elective cholecystectomy, elective bowel resection, hysterectomy	3-10 %
III - Contaminated	Fresh accidental wound; major break in technique ; gross GI spillage; acute non-purulent inflammation	Fresh trauma, enterotomy with spillage	15-20 %
IV - Dirty / infected	Old wound with devitalised tissue; existing infection or perforated viscus	Faecal peritonitis, perforated appendix	30-40 %

- **CDC depth classification:** **superficial incisional** (skin/subcutaneous, within **30 days**); **deep incisional** (fascia/muscle, within **30 days, or 90 days if an implant** is present); **organ/space**.
- Commonest organism = **Staphylococcus aureus**; commonest **source of contamination = the patient's own endogenous flora**.
- **Proven SSI-prevention bundle:** CHG-alcohol prep, no razor shaving, antibiotic within 60 min, **normothermia > 36 °C, glucose < 180 mg/dL**, adequate tissue oxygenation, minimal traffic, good haemostasis and no dead space.

6. ELECTRO SURGERY

The PDF gives 5 precautions. The mechanism and the injury mechanisms are what carry the marks.

6.1 Fundamentals

Concept	Detail
Principle	High-frequency alternating current concentrated at a small active electrode; tissue resistance converts it to heat → cutting, coagulation or desiccation . It is not cautery (true cautery = a passively heated wire, no current through the patient)
Frequency	300 kHz – 3 MHz (radiofrequency) . Above ~ 100 kHz the current does not stimulate nerve or muscle (the Faradic effect is avoided) — which is why RF current cuts but does not electrocute
Cut waveform	Continuous, low-voltage, high-current (100 % duty cycle) → vaporisation, little haemostasis
Coag waveform	Interrupted, high-voltage, low duty cycle (~6 %) → coagulum. High voltage → greater risk of capacitive coupling, insulation breakdown and sparking
Monopolar circuit	Generator → active electrode (pencil) → patient → dispersive/return pad → generator. Current crosses the whole body; the pad is essential
Bipolar circuit	Both electrodes are the two tines of the forceps ; current passes only between them. No pad needed , minimal lateral spread, usable in fluid, safest with pacemakers — neurosurgery, ophthalmic, microsurgery
Pad placement	Clean, dry, hair-free, well-vascularised muscle mass, as close as practicable to the site , whole surface in uniform contact, away from ECG electrodes , bony prominences, scar, tattoos, metal implants and pooled fluid
REM / contact-quality monitoring	A split pad whose two halves are monitored for impedance; the generator shuts down if contact becomes inadequate — has virtually eliminated pad-site burns. Modern generators are isolated , not ground-referenced
AEM (active-electrode monitoring)	Shielded laparoscopic instruments that drain stray current — prevents insulation-failure and capacitive-coupling burns

6.2 The five mechanisms of electrosurgical injury

- 1. **Return-electrode (pad-site) burn** — poor or partial pad contact, hair, dried gel, pad over bone or scar. Prevented by correct placement + REM.
- 2. **Alternate-site burn** — the patient's skin touches a **metal surface** (table edge, stirrup, armboard screw) which becomes an unintended return path. The PDF states this rule twice: **no skin-to-metal contact**.
- 3. **Direct coupling** — the activated electrode touches another metal instrument (laparoscope, retractor), which burns adjacent tissue.
- 4. **Capacitive coupling** — current crosses **intact insulation** by an electrostatic field; classic setting = a **metal instrument through a plastic (hybrid) trocar**, or coiled cords. Worse with **coag mode and long thin instruments**.
- 5. **Insulation failure** — a pin-hole break delivers current **outside the surgeon's view** → **delayed bowel perforation 3-10 days post-operatively**. Inspect insulation before every use.

6.3 Other examination points

- **Repeated requests to increase the power** = poor pad contact, a loose connection, a frayed cord or a failing generator. **Stop, check the circuit and pad; do not simply turn it up**. Replace and label the unit and **document the serial numbers and the reason**.
- **Keep the pencil in its holster** when not in use; clean eschar with a non-abrasive pad.
- **Pacemaker / ICD**: prefer **bipolar**; if monopolar is essential use short bursts at low power with the current path away from the device, keep the **crash cart with defibrillator immediately available and document that it is there**, monitor ECG and pulse continuously, and **interrogate the device afterwards**.
- **Never use ESU or laser in an oxygen-enriched field** (mouth, oropharynx, tracheostomy, under drapes with high-flow O₂) — pause the oxygen and allow washout.
- **Advanced devices in one line each: ultrasonic (Harmonic) scalpel** — mechanical vibration at **55.5 kHz**, cavitation at a **lower temperature (50-100 °C)**, no current through the patient; **bipolar vessel sealing (LigaSure)** — seals vessels up to **7 mm**; **argon-enhanced** coagulation — thin uniform eschar, gas-embolism risk.
- **Surgical smoke (plume)**: > 150 chemicals plus viable cells and viral DNA (**documented HPV transmission to surgeons**). **Ablating 1 g of tissue ≈ 3-6 unfiltered cigarettes**. Control = **smoke evacuator with an ULPA filter (99.999 % at 0.1 μm) held within 2.5-5 cm (1-2 in)** plus a **high-filtration mask or N95** — a standard mask is **not** adequate.

EQUIPMENT NUMBERS THAT SHARE THIS SUB-TOPIC

Pneumatic tourniquet: cuff ends overlap **3-6 inches**; **never at or near the elbow**; **upper limb 250-300 mmHg for ≤ 60 min**, **lower limb 300-350 mmHg for ≤ 90-120 min**; **deflate 10-15 min** if longer; exsanguinate with an Esmarch bandage **before** inflation; **record site, pressure and both inflation and deflation times**. On release: ↓ BP, transient acidosis, ↑ K⁺, ↑ EtCO₂.

Laser: **wavelength-specific eye protection for everyone including the patient**; warning signs on doors, windows covered, key held by the laser safety officer, **standby mode when not firing, foot pedal only to the surgeon, matte instruments, wet drapes, saline available, plume evacuator, laser-safe ET tube with a saline-filled cuff, and never in an oxygen-enriched field**.

Powered saws/drills: engineering clearance, test before use, attachments secure and sharp, **let the rotating part come to a complete standstill before setting it down**, face shield mandatory.

7. OPERATIVE RECORDS

7.1 Specimen handling — a favourite MCQ the PDF omits

Specimen	Handling
Routine histopathology	10 % neutral buffered formalin , volume ≈ 10× the specimen ; container labelled with name, ID, date, site/side; label read aloud during Sign Out
Frozen section	FRESH, in saline-moistened gauze — NEVER in formalin (formalin prevents further study and immunohistochemistry); send immediately
Culture / microbiology	Sterile container, no fixative ; anaerobic transport for anaerobes; never formalin
Stones/calculi	Dry container, no fixative
Implants	Record type, manufacturer, size, serial/lot number in the intra-operative record and the implant log (traceability for recalls)
Lost specimen	A serious, irreplaceable incident — incident report mandatory

Q9. The legal responsibility for recording the surgical count rests with the
(a) Surgeon (b) Scrub person (c) Circulator (d) Anaesthetist
Ans: (c) Circulator — Both count **aloud together** as the scrub person touches each item, but **keeping the written record is the circulator's legal responsibility**, and the circulator initials every addition.

B — Transfer and Positioning

Q10. The single most important measure to prevent a patient falling from the operating table is
(a) Side rails (b) A safety strap firmly applied across the mid-thighs (c) Padded armboards (d) Two staff at the bedside
Ans: (b) A safety strap firmly applied across the mid-thighs — The strap across the **mid-thighs, ≈ 3 inches (7.5 cm) above the knees**, firm but not tight enough to impair venous return. The patient is also **never left unattended** in the OR.

Q11. The nerve most commonly injured in the lithotomy position is the
(a) Femoral nerve (b) Sciatic nerve (c) Common peroneal nerve (d) Obturator nerve
Ans: (c) Common peroneal nerve — The **common peroneal nerve** is compressed against the **fibular head** by the stirrup post → **foot drop** and loss of dorsal foot sensation. Lithotomy also risks **lower-limb compartment syndrome** if prolonged.

Q12. The commonest peripheral nerve injury in the supine position involves the
(a) Radial nerve (b) Ulnar nerve (c) Median nerve (d) Axillary nerve
Ans: (b) Ulnar nerve — **Ulnar neuropathy** is the commonest peri-operative nerve injury overall. Prevent by abducting the arms **≤ 90°**, padding the elbows and placing the **palms up (supinated)**.

Q13. All of the following occur in the Trendelenburg position EXCEPT
(a) Increased intracranial pressure (b) Increased venous return (c) Increased functional residual capacity (d) Cephalad displacement of the diaphragm
Ans: (c) Increased functional residual capacity — Trendelenburg **decreases** FRC and compliance because abdominal contents push the diaphragm cephalad → atelectasis. It increases venous return, CVP, **ICP and IOP**, and risks facial/laryngeal oedema and tube migration.

Q14. Reverse Trendelenburg position is typically used for
(a) Abdominal hysterectomy (b) Thyroidectomy (c) Haemorrhoidectomy (d) Nephrectomy
Ans: (b) Thyroidectomy — Reverse Trendelenburg (head up) is used for **neck surgery — thyroidectomy, parathyroidectomy, scalene node biopsy — and laparoscopic cholecystectomy**. Trendelenburg is for pelvic surgery, lithotomy/jackknife for anorectal, lateral kidney for nephrectomy.

Q15. The Kraske (jackknife) position is a modification of the
(a) Supine position (b) Prone position (c) Lateral position (d) Fowler's position
Ans: (b) Prone position — **Kraske/jackknife is a PRONE modification** exposing the sacrococcygeal and anorectal area. **Trendelenburg, reverse Trendelenburg, Fowler's and lithotomy are all modifications of SUPINE.**

Q16. Both legs must be lowered slowly and simultaneously from the lithotomy position mainly to prevent
(a) Nerve traction (b) A sudden fall in blood pressure (c) Wound dehiscence (d) Deep vein thrombosis
Ans: (b) A sudden fall in blood pressure — Lowering the legs shifts blood into the lower limbs → **sudden reduction in venous return and a precipitous fall in blood pressure**. Two persons lower them together, slowly.

Element	Detail
Oxidiser	Oxygen and nitrous oxide — O ₂ pools under drapes near the head (the classic head-and-neck fire)
Prevention	Let alcohol dry ≥ 3 min , never allow pooling, blot puddles; keep FiO ₂ < 30 % for head/neck work; pause oxygen before activating ESU/laser in the airway ; moisten sponges and drapes; holster the pencil
Response — R A C E	Rescue • Alarm (call the code, e.g. CODE RED) • Confine (close doors, shut off medical gases) • Extinguish/Evacuate
Extinguisher — P A S S	P ull the pin • A im at the base of the fire • S queeze • S weep
Extinguisher classes	A ordinary combustibles • B flammable liquids • C electrical → CO₂ or dry chemical, NEVER water • D metals • K kitchen fats. The OR stocks CO₂ or ABC dry chemical ; CO ₂ is preferred around equipment (no residue)
Airway fire drill	Stop ventilation and disconnect the oxygen → REMOVE the ET tube and burning material → pour saline into the airway → ventilate on air → re-intubate and bronchoscope → steroids/ICU ; then incident report and device reporting
Drills	At least twice a year, unannounced . Everyone must know the code name and the location of the alarm, extinguishers, gas shut-off valves and evacuation route. Drapes and gowns are fire-RETARDANT, not fire-PROOF

10.5 Disaster preparedness and triage

Concept	Content
Types	Internal — fire, power/gas failure, structural, violent intruder; External — earthquake, flood, road/rail crash, blast, chemical leak, epidemic
Framework	A written Hospital Disaster Management Plan based on a Hazard Vulnerability Analysis , run through a Hospital Incident Command System , with call-tree, surge capacity and mock drills at least annually
START triage	Simple Triage And Rapid Treatment — ≤ 60 s per casualty , assessing RPM: Respiration, Perfusion, Mental status
Triage colours	RED = Priority I, immediate (RR > 30, absent radial pulse / capillary refill > 2 s, or not obeying commands — airway obstruction, major haemorrhage, shock) • YELLOW = Priority II, delayed (can wait 1–4 h) • GREEN = Priority III, minor / walking wounded • BLACK = Priority 0, dead or expectant (not breathing after the airway is opened)
Key principle	In a mass casualty the aim shifts from ‘the best for each individual’ to ‘the greatest good for the greatest number’ ; triage is dynamic — re-triage at every stage
Power failure	Automatic emergency generator back-up, UPS for monitors and ventilators; everyone must know where the high-intensity torches are kept, and where the self-inflating bag and reserve oxygen cylinder are for medical-gas failure
Emergency signal	Each OR has a special emergency signal (usually a flashing red light) operable from within that room
Code colours (intitution-specific)	Typically Blue cardiac arrest • Red fire • Pink infant abduction • Orange mass casualty • Black bomb threat • Brown evacuation • Grey violent person. Answer per your own institution's list
Pharmacist's disaster role	Maintain and rotate disaster drug stock and antidote kits (atropine–pralidoxime, potassium iodide, naloxone, antivenom, tetanus toxoid), bulk-prepare fluids and analgesics, control narcotic documentation, advise on dosing in renal failure/crush injury, manage the cold chain

11. IN-SERVICE EDUCATION, BODY MECHANICS AND FATIGUE FACTORS

The PDF covers all three well. Only the quantitative anchors are missing.

- **Orientation** = initial familiarisation with policies and the environment (**AORN recommends a mentor / buddy system**). **In-service education** = ongoing, employer-provided, job-specific — perioperative departments hold it **monthly**. **Continuing education** = professional development beyond the employer.
- **Mandatory recurring training:** fire safety with **drills at least twice a year (unannounced)**, disaster drill annually, infection control and hand hygiene, biomedical waste, needle-stick protocol, radiation safety, laser safety, electrical safety, hazard communication, **BLS recertification every 2 years**, ergonomics, and **competency validation for every new device before use**.
- **Body mechanics:** lift by **bending the knees and hips with the back straight**, position your body under the load, hold it close, feet apart, **straighten the legs so the lower-limb muscles do the work, pivot with the feet — never twist the spine**, push rather than pull, lift in unison on a count.
- **NIOSH maximum recommended load = 51 lb (23 kg); patient-handling limit 35 lb (16 kg)**. OSHA (2002) requires facilities to reduce ergonomic injury; **lift teams** are cost-effective because back injury is the commonest occupational injury in health care.
- **Fatigue:** AORN recommends **≤ 12 consecutive hours** of direct patient care, **≤ 60 h/week including on-call**, and **8 h of uninterrupted sleep**. Being awake **17-19 hours** impairs performance about as much as a blood alcohol of 0.05 %. **Working with impaired judgement is an ethical breach, not just a personal risk**.
- **Other occupational hazards:** prolonged standing (varicose veins), repetitive strain in laparoscopy, **noise > 85 dB** from saws and drills, sharps injury, latex, and burnout.

12. RADIATION SAFETY

12.1 Units

Quantity	SI unit	Old unit	Conversion
Absorbed dose	Gray (Gy) = 1 J/kg	rad	1 Gy = 100 rad
Equivalent / effective dose	Sievert (Sv)	rem	1 Sv = 100 rem — the unit used for all dose limits and personnel monitoring
Activity	Becquerel (Bq)	Curie (Ci)	1 Ci = 3.7 × 10 ¹⁰ Bq
Exposure	Coulomb/kg	Roentgen (R)	Ionisation in air

12.2 Dose limits (ICRP / AERB)

Category	Limit
Occupational, whole body	20 mSv/year averaged over 5 years (100 mSv/5 y), maximum 50 mSv in any one year
Occupational, lens of the eye	20 mSv/year (ICRP 2011 revision; formerly 150 mSv)
Occupational, skin / extremities	500 mSv/year
Declared pregnancy	≤ 1 mSv to the fetus for the remainder of the pregnancy (0.5 mSv/month)
Public	1 mSv/year

A — Pre-Operative Preparation, Identification and Records

Q1. The minimum fasting period for clear liquids before elective surgery in a healthy adult is

- (a) 1 hour (b) 2 hours (c) 4 hours (d) 6 hours

Ans: (b) 2 hours — Remember 2-4-6-8: clear fluids 2 h, breast milk 4 h, formula/light meal 6 h, fatty meal 8 h. The purpose is to prevent **aspiration (Mendelson's syndrome)**; prolonged fasting is harmful and does not reduce gastric volume.

Q2. The preferred method of pre-operative hair removal is

- (a) Shaving with a razor the night before (b) Shaving with a razor in the OR (c) Electric clippers immediately before surgery (d) Hair is always left intact

Ans: (c) Electric clippers immediately before surgery — Ideally hair is **not removed**; if it interferes, use **clippers immediately before surgery, outside the OR**. Razor shaving — especially the night before — creates micro-abrasions and carries the **highest** SSI risk.

Q3. The WHO Surgical Safety Checklist consists of how many phases?

- (a) Two (b) Three (c) Four (d) Five

Ans: (b) Three — **Three: Sign In** (before induction), **Time Out** (after induction, before incision), **Sign Out** (before leaving the OR) — 19 items. It reduced mortality from 1.5 % to 0.8 %.

Q4. The surgical “time out” must be performed

- (a) On admission to the ward (b) In the holding area (c) After induction but before skin incision (d) Immediately after wound closure

Ans: (c) After induction but before skin incision — The time out is the whole-team verbal pause **immediately before incision**. The **Universal Protocol** was mandated by the Joint Commission in **July 2004**; **AORN added the correct patient position in March 2005**.

Q5. Which of the following is verified during the “Sign Out” phase?

- (a) Pulse oximeter functioning (b) Site marking (c) Correctness of instrument, sponge and needle counts (d) Difficult airway risk

Ans: (c) Correctness of instrument, sponge and needle counts — Sign Out covers the **name of the procedure recorded, correctness of counts, specimen labelling read aloud, equipment problems and key recovery concerns**. Pulse oximetry, site marking and airway risk belong to **Sign In**.

Q6. A specimen for frozen section must be sent

- (a) In 10 % formalin (b) Fresh, in saline-moistened gauze (c) In absolute alcohol (d) In 2 % glutaraldehyde
- Ans: (b) Fresh, in saline-moistened gauze — **Frozen sections go FRESH — never in formalin**, which prevents further processing and immunohistochemistry. Routine histopathology goes in **10 % neutral buffered formalin at ≈ 10× the specimen volume**; culture specimens go in a sterile container with no fixative.

Q7. Consent for surgery on a 10-year-old child is governed by which section of the IPC?

- (a) Section 87 (b) Section 88 (c) Section 89 (d) Section 92

Ans: (c) Section 89 — § 89 — acts done in good faith for a **child under 12** or a person of unsound mind, with the guardian's consent. **§ 90** — consent given under fear or misconception is invalid. **§ 92** — acts in good faith **without** consent in an emergency.

Q8. A retained surgical sponge presenting years later as an abdominal mass is called a

- (a) Granuloma annulare (b) Gossypiboma (c) Pseudocyst (d) Phlegmon

Ans: (b) Gossypiboma — **Gossypiboma (textiloma)** — gauze encased in a foreign-body granuloma. The sponge is the **commonest retained item (50-70 %)** and this is a classic **res ipsa loquitur** claim. Risk factors: emergency surgery, **unexpected change of procedure** and high BMI.

Topic	Number
Biological indicators	Steam & plasma → Geobacillus stearothermophilus • dry heat & EO → Bacillus atrophaeus • radiation → Bacillus pumilus
Electrosurgery	300 kHz - 3 MHz ; no nerve stimulation above ~100 kHz ; ultrasonic scalpel 55.5 kHz ; LigaSure seals 7 mm ; smoke — 1 g tissue $\approx$ 3-6 cigarettes , evacuator within 1-2 in, ULPA 99.999 % at 0.1 μm
Tourniquet	Upper limb 250-300 mmHg / $\leq$ 60 min • lower limb 300-350 mmHg / $\leq$ 90-120 min ; overlap 3-6 in ; deflate 10-15 min if longer
CPR	100-120/min • 5-6 cm • 30:2 • biphasic 120-200 J • adrenaline 1 mg q3-5 min • amiodarone 300 mg
Anaphylaxis	Adrenaline 0.5 mg IM (1:1000) , repeat q5 min; IV 50 μg boluses; fluid 20 mL/kg
Malignant hyperthermia	Dantrolene 2.5 mg/kg up to 10 (-30) mg/kg • O₂ 10 L/min • cool to 38 °C • urine > 1-2 mL/kg/h • RYR1, chromosome 19
Radiation	1 Gy = 100 rad • 1 Sv = 100 rem ; occupational 20 mSv/y (max 50 in one year) ; public 1 mSv/y ; eye 20 mSv/y ; extremities 500 mSv/y ; fetus 1 mSv (0.5 mSv/month); apron 0.25-0.5 mm Pb ; stand $\geq$ 2 m ; doubling distance $\rightarrow$ $\frac{1}{4}$
Needle-stick	HBV 6-30 % • HCV 1.8 % • HIV 0.3 % ; HIV PEP within 2 h (max 72 h) for 28 days ; HBIG 0.06 mL/kg within 24 h
Chemical limits	EO 1 ppm • formaldehyde 0.75 ppm • glutaraldehyde 0.05 ppm ceiling • N₂O 25 ppm • volatiles 2 ppm • methyl methacrylate 100 ppm • noise 85 dB
Lifting	NIOSH 51 lb (23 kg) ; patient handling 35 lb (16 kg)
Waste	Yellow anatomical/soiled/expired drugs/cytotoxic • Red recyclable plastics • White sharps • Blue glass and metallic implants; bags to $\frac{3}{4}$; cytotoxic incineration $\geq$ 1200 °C ; storage $\leq$ 48 h
Fatigue & drills	$\leq$ 12 consecutive hours • $\leq$ 60 h/week • 8 h sleep ; fire drills $\geq$ 2/year ; BLS recertification every 2 years
Recovery scores	Aldrete max 10, discharge $\geq$ 9 (A-R-C-C-O) • PADSS $\geq$ 9 • Apfel 4 factors $\rightarrow$ 10/20/40/60/80 %
Post-op timeline	Fever: Wind 1-2 • Water 3-5 • Walking 4-6 • Wound 5-7 • Drugs > 7 • dehiscence POD 5-10 • secondary haemorrhage POD 7-14 • DVT POD 5-10
Wound healing	Inflammatory 0-4 d • proliferative 4-21 d • remodelling 21 d-1 y ; tensile strength 5-10 % at 1 wk, 30-50 % at 1 month, max 70-80 % ; sutures out face 3-5, trunk 7-10, limbs 10-14, joints 14 d
Bowel & urine	Small bowel 24 h $\rightarrow$ stomach 24-48 h $\rightarrow$ colon 48-72 h ; urine output > 0.5 mL/kg/h

17. MCQ BANK — 77 QUESTIONS, SYLLABUS-WEIGHTED

ABOUT THESE QUESTIONS

77 questions, every one mapped to a listed syllabus sub-point. Questions on off-syllabus topics (ASA grading, premedication pharmacology, analgesic choice, fluid regimens) have been removed from the earlier draft. They are **modelled on the pattern and difficulty of previous-year questions** in Indian pharmacy, nursing and OT-technology papers, and **written from the standard sources** — AORN Guidelines, Berry & Kohn's Operating Room Technique, CDC/HICPAC guidelines, WHO, ICRP/AERB, OSHA/NIOSH, BMW Rules 2016 and Bailey & Love. They are **not verbatim reproductions** of any single paper.

Weighting, matched to where the marks fall: **sterilization 14 • post-operative care 13 • environment/asepsis/SSI 10 • electrosurgery & equipment 9 • pre-op preparation, identification, records & counting 9 • positioning 8 • emergencies 7 • radiation, chemical hazards & waste 7.**

12.3 Protection — three cardinal principles under ALARA

Principle	Application
1. Time	Minimise screening time; pulsed fluoroscopy, last-image hold, collimation ; record total screening time
2. Distance — INVERSE SQUARE LAW	Intensity $\propto$ 1/d² , so doubling the distance gives $\frac{1}{4}$ the exposure and tripling gives $\frac{1}{9}$ ² = $\frac{1}{9}$. Stand $\geq$ 2 m (6 ft) away or leave the room; in lateral views stand on the image-intensifier side, away from the tube (scatter is highest on the tube side)
3. Shielding	Lead apron 0.25-0.5 mm lead equivalence (0.5 mm attenuates $\approx$ 95-99 % at 100 kVp), thyroid collar, lead glasses, lead gloves for holding cassettes, mobile lead screens . Aprons are hung, never folded , and screened annually for cracks
ALARA	As Low As Reasonably Achievable — with ICRP's justification, optimisation and dose limitation
Monitoring	Dosimeter worn OUTSIDE the lead apron at chest/collar level (a second one inside at waist level in pregnancy). TLD is commonest; also film badge, OSL, electronic personal dosimeter. Records kept for life
Personnel rules	Non-essential staff leave the room or wear a lead apron ; scrubbed staff don the apron BEFORE scrubbing ; never hand-hold the cassette or patient if avoidable; pregnant staff avoid all radiation cases
Patient rules	Ask every woman of child-bearing age if she is or could be pregnant ; shield gonads and thyroid when the beam cannot be collimated; lowest diagnostic exposure
Radioactive implants	Handle with long forceps, never fingers ; lead pot in the room; limit and rotate staff time; exclude pregnant staff and children; survey the room and linen with a Geiger-Müller counter afterwards

12.4 Biological effects

Effect	Nature	Examples
Deterministic (non-stochastic)	Has a threshold ; severity increases with dose	Skin erythema, epilation, cataract , sterility, marrow suppression, fetal malformation
Stochastic	No threshold ; probability increases with dose (linear-no-threshold)	Carcinogenesis and heritable mutation

Most radiosensitive: bone marrow and lymphoid tissue, gonads, intestinal crypts, skin basal layer, lens, and the fetus (most sensitive in weeks 2-8, organogenesis). Least sensitive: adult nerve, muscle and bone.

13. INFECTION CONTROL

13.1 Standard vs transmission-based precautions

Category	Applies to	Requirements	Examples
Standard Precautions (CDC 1996; absorbed the older Universal Precautions)	Every patient, every time — all blood, body fluids, secretions and excretions except sweat , non-intact skin, mucous membranes	Hand hygiene • gloves • gown • mask/eye protection when splash is likely • safe injection and sharps practice • respiratory hygiene • safe handling of equipment, linen and waste • environmental cleaning	Universal application in the OR

Category	Applies to	Requirements	Examples
Contact	Direct/indirect contact	Standard + gown and gloves for all contact , dedicated equipment, single room preferred	MRSA, VRE, CRE, C. difficile (soap and water; hypochlorite for surfaces), scabies, RSV
Droplet	Large droplets > 5 µm travelling ≈ 1-2 m (3-6 ft)	Standard + surgical mask within 1-2 m , single room, mask the patient during transport	Influenza, pertussis, mumps, rubella, N. meningitidis, diphtheria
Airborne	Droplet nuclei ≤ 5 µm that stay suspended	Standard + fit-tested N95 , and an Airborne Infection Isolation Room — NEGATIVE pressure with ≥ 12 air changes/hour , door closed	Pulmonary/laryngeal TB, measles, varicella

13.2 PPE sequence — rote-learn the order

Donning (put on)	Doffing (take off)
1 Hand hygiene → 2 Gown → 3 Mask/respirator (fit-check) → 4 Goggles/face shield → 5 Gloves	1 Gloves → 2 Goggles/face shield → 3 Gown → 4 Mask/respirator (remove outside the room, by the ties only) → 5 Hand hygiene

The most contaminated item comes off first (gloves) and the respirator comes off last, because it protects you until you have left the room.

13.3 Sharps injury and post-exposure prophylaxis

Item	Detail
Transmission risk per percutaneous needle-stick	Hepatitis B 6-30 % (up to 30 % if HBeAg-positive — the highest) • Hepatitis C ≈ 1.8 % • HIV ≈ 0.3 % (1 in 300) ; mucous-membrane HIV exposure ≈ 0.09 %
Immediate first aid	Do NOT squeeze, suck, or apply bleach/alcohol/iodine into the wound. Wash with soap and running water ; irrigate eye/mucosa with water or saline for 10-15 min ; allow to bleed, dry and cover
Then	Report immediately → assess the exposure → baseline serology of both parties → start PEP → counselling → follow-up at 6 weeks, 3 months, 6 months → document and file an incident report
HIV PEP	As early as possible — ideally within 2 hours, never later than 72 hours; 3-drug regimen for 28 days
Hepatitis B PEP	Unvaccinated or non-responder → HBIG 0.06 mL/kg IM within 24 hours plus start the vaccine course. A vaccinated responder (anti-HBs ≥ 10 mIU/mL) needs nothing . HBV vaccination (0, 1, 6 months) is mandatory for health-care workers
Hepatitis C	No vaccine and no PEP — monitor and treat early infection with direct-acting antivirals

MEDICATION SAFETY IN THE OR — the pharmacist's angle on this unit Extend the PDF's five rights to the eight rights: right patient, drug, dose, route, time, plus right documentation, indication and response. Every medication on the sterile field is labelled immediately with name, strength and concentration; unlabelled medications are ALWAYS discarded; all vials and syringes stay in the room until the end of the case; labelled medications are shown to the relief person at hand-over; calculations are independently double-checked by a second licensed professional. High-alert drugs in the OR: concentrated potassium chloride, insulin, heparin, neuromuscular blockers ('paralysing agent — warning' label), concentrated adrenaline, local anaesthetics, oxytocin, and look-alike/sound-alike pairs (ephedrine/epinephrine). Constraints recommended by AORN: minimise the number of calculations required, stock dose-specific ready-to-use concentrations, standardise concentrations, use pre-filled syringes, and assess competency regularly. Also the pharmacist's OT duties: crash-cart stock, expiry and seal checks, cold-chain logs, narcotic accounting (NDPS register), sterile dilution of injectables, ADR reporting (pharmacovigilance) and in-service education on new drugs and devices.

16. THE NUMBERS SHEET

Topic	Number
Fasting	2 h clear fluids • 4 h breast milk • 6 h formula/light meal • 8 h fatty meal ; aspiration risk if gastric volume > 25 mL and pH < 2.5
Antibiotic prophylaxis	Within 60 min of incision ; re-dose after 2 half-lives or > 1500 mL blood loss; stop within 24 h
Identification	≥ 2 identifiers ; WHO checklist = 3 phases, 19 items ; Universal Protocol July 2004 ; AORN position addition March 2005
Safety strap	Mid-thighs, ≈ 3 inches (7.5 cm) above the knees ; arms abducted ≤ 90° ; Trendelenburg 30-45°
Sterile field	Wrapper edge unsterile 2.5 cm (1 in) • unsterile person keeps 30 cm (1 ft) away • gown sleeves sterile to 5 cm above the elbow
Scrub	First scrub of the day 3-5 min ; double gloving cuts perforation 70-80 %
OR environment	20-23 °C • RH 20-60 % • ≥ 15 (ideally 20-25) air changes/h • POSITIVE pressure • HEPA 99.97 % at 0.3 µm • laminar flow 300-600 ACH • noise < 45 dB
Wound class SSI risk	Clean < 2 % • clean-contaminated 3-10 % • contaminated 15-20 % • dirty 30-40 % ; SSI window 30 days (90 with an implant)
Autoclave	121 °C / 15 psi / 15-20 min • 134 °C / 30 psi / 3-3.5 min • prevacuum 132-135 °C / 3-4 min • flash 132 °C / 3 min unwrapped (10 min lumened) • prions 134 °C / 18 min
Hot air oven	160 °C / 2 h • 170 °C / 1 h • 180 °C / 30 min
Bowie-Dick	134 °C for 3.5 min in an EMPTY prevacuum chamber, daily — tests air removal
Ethylene oxide	450-1200 mg/L • 37-63 °C • RH 40-80 % • 1-6 h ; aeration 8-12 h at 50-60 °C or 7 days at room temperature
H ₂ O ₂ plasma	58 % H₂O₂ • 45-50 °C • ≈ 1 h • no cellulose, no liquids
Liquid chemicals	Glutaraldehyde 2 % — 10 h sterilization / 20-45 min HLD • OPA 0.55 % — 12 min • peracetic acid — 12 min at 50-56 °C • H₂O₂ 7.5 % — 30 min HLD
Sterility	SAL 10⁻⁶ • γ-radiation 25 kGy • filter 0.22 µm • shelf-life is event-related

Drain	Type	Key point
Chest (intercostal) drain	Closed, underwater seal	Keep the bottle below chest level; never clamp a bubbling drain; ‘swinging’ = patent, continuous bubbling = air leak. If disconnected → place the end in sterile water; if it falls out → occlusive dressing taped on three sides
Nasogastric tube	Decompression / feeding	Confirm position by pH < 5.5 on aspirate or X-ray — the auscultation ‘whoosh’ test is unreliable and unsafe
Urinary catheter	Closed drainage	Closed system, bag below bladder level and off the floor, no dependent loops, remove within 24 h. CAUTI is the commonest healthcare-associated infection

- **Drain care:** secure against traction, keep patent, chart **volume, colour and character** each shift, aseptic site dressing, and **never advance a drain back in. Brisk bright-red output → haemorrhage; faecal or bilious output → anastomotic leak; cloudy or foul → infection** — report immediately.
- **Bowel function returns in the order: small intestine ≈ 24 h → stomach 24-48 h → COLON 48-72 h (last).** Resolution is judged by the **passage of flatus or stool**, not bowel sounds. **Ileus** — absent bowel sounds and generalised distension; **mechanical obstruction** — colicky pain with high-pitched tinkling sounds.
- **Urine output must exceed 0.5 mL/kg/h (≈ 30 mL/h).** Post-operative oliguria is **most often hypovolaemia; urinary retention** (commonest after anorectal, hernia, pelvic and spinal-anaesthetic surgery) shows as suprapubic fullness, restlessness and overflow dribbling → privacy, upright position, warm compress, analgesia, then **in-and-out catheterisation**.
- **DVT peaks POD 5-10** and is often silent; **PE** presents as sudden dyspnoea, pleuritic pain, tachycardia and hypoxia, and is a leading cause of **preventable** post-operative death. Prevention = **early ambulation, ankle exercises, TED stockings, intermittent pneumatic compression, LMWH, hydration and adequate analgesia so the patient can move.**
- **Respiratory bundle:** semi-Fowler's position, **incentive spirometry 10 breaths hourly, splinted coughing**, adequate analgesia, **ambulation within 24 h**, oral hygiene, humidified oxygen.
- **Red flags to teach the patient before discharge:** fever, increasing pain, spreading redness, purulent or foul discharge, wound gaping, bleeding, calf pain or swelling, chest pain or breathlessness, persistent vomiting, inability to pass urine.

Item	Detail
Prevention hierarchy	Elimination/substitution (needleless systems) → engineering controls (retractable/sheathed needles, blunt suture needles, sharps container filled to ¾) → work-practice controls (never recap by hand — if unavoidable use the one-handed ‘scoop’; never pass a sharp hand-to-hand — use a neutral zone / hands-free technique with a magnetic pad; mount blades with a needle holder) → PPE (double gloving cuts inner-glove perforation by 70-80 %, face shield)

13.4 Biomedical waste segregation — BMW Rules 2016 (India)

Colour	Contents	Treatment / disposal
YELLOW	Human and animal anatomical waste , soiled waste (dressings, cotton, blood-soaked items, plaster casts), expired and discarded medicines , chemical waste, contaminated linen, microbiology and laboratory waste, and cytotoxic waste in a separate yellow ‘cytotoxic’ container	Incineration / plasma pyrolysis / deep burial (deep burial only in towns < 500 000). Cytotoxic waste → incineration at ≥ 1200 °C or return to the manufacturer
RED	Contaminated recyclable plastics — IV tubing and sets, catheters, urine bags, syringes without needles , gloves	Autoclave / microwave / hydroclave, then shred → registered recycler. No chemical pre-treatment, no landfilling
WHITE (translucent, puncture- and leak-proof)	Waste sharps including metals — needles, syringes with fixed needles , scalpels, blades	Autoclave or dry-heat sterilise, then shred/mutilate; final disposal by encapsulation in a concrete sharps pit or to an iron foundry
BLUE	Glassware (broken/discarded contaminated glass, medicine vials and ampoules) and metallic body implants	Disinfection (1 % hypochlorite) or autoclaving, then recycling

- **Golden rule: segregate at the point of generation, by the person generating the waste.** Bags filled to ¾, tied (not stapled), labelled with the **biohazard/cytotoxic symbol**, stored **≤ 48 hours**.
- **Never recap, bend or break needles by hand** — use a needle-tip cutter at the point of use; the needle goes to the **white** container.
- **Glass vials contaminated with cytotoxics go to YELLOW, not blue** — a classic trick question.
- **Mercury spill:** do **not** vacuum; collect with a syringe/adhesive tape or a spill kit into a sealed container of water and dispose as hazardous waste.
- Requirements: annual report to the Pollution Control Board, waste register, **annual health check-up and immunisation (hepatitis B, tetanus) of waste handlers**, PPE, and reporting of any major accident within 24 h. **Radioactive waste is governed separately by the AERB** (decay-in-storage).

14. CHEMICAL HAZARDS

14.1 Occupational exposure limits worth memorising

Agent	Limit	Effects and control
Nitrous oxide	NIOSH REL 25 ppm	Megaloblastic change, neuropathy, ↑ spontaneous abortion and reduced fertility . Control: scavenging system, no leaks, tight masks, adequate air changes

Agent	Limit	Effects and control
Halogenated volatiles	NIOSH REL 2 ppm (0.5 ppm with N ₂ O)	Headache, fatigue, impaired mental performance, audiovisual ability and manual dexterity ; hepatotoxicity; teratogenic concern
Ethylene oxide	OSHA PEL 1 ppm; excursion 5 ppm/15 min	Irritant, mutagen, carcinogen, flammable/explosive . Dedicated ventilated room, gas monitors, aeration cabinet, personal badges
Formaldehyde	OSHA PEL 0.75 ppm; STEL 2 ppm	Irritation, asthma, dermatitis, Group 1 carcinogen — why OT fumigation is abandoned. Handle specimens in a ventilated hood
Glutaraldehyde	ACGIH ceiling 0.05 ppm	Occupational asthma, rhinitis, conjunctivitis, contact dermatitis, epistaxis . Use a covered basin with local exhaust, nitrile or butyl gloves (NOT latex — it permeates), goggles and apron
Methyl methacrylate (bone cement)	OSHA PEL 100 ppm	Irritant, dermatitis, headache; and in the patient cement implantation syndrome — hypotension, hypoxia, arrhythmia, arrest during cementing. Mix in a closed vacuum system with local exhaust
Noise	OSHA action level 85 dB (8-h TWA)	Saws, drills, suction, alarms — hearing protection; also impairs communication
Compressed gases	—	Secure every cylinder upright; never drop one or let cylinders strike each other; valve cap on for transport; never oil or grease an oxygen valve; segregate full and empty. A falling cylinder becomes a missile

14.2 Hazard communication, cytotoxics and spills

- **Hazard Communication / ‘Right to Know’:** every hazardous chemical needs a **Safety Data Sheet (SDS, formerly MSDS) in 16 standardised sections**, accessible to all staff at all times, plus **GHS labelling** with the signal word **‘Danger’** or **‘Warning’** and the nine pictograms.
- **Hierarchy of controls, in order:** **elimination → substitution → engineering controls (ventilation, closed systems, scavenging) → administrative / work-practice controls → PPE (the LAST and weakest line)**. The ranking itself is commonly asked.
- **Cytotoxic handling:** prepare in a **Class II Type B biological safety cabinet or containment isolator in a negatively-pressured, externally-vented room**; use **closed-system transfer devices** and Luer-lock syringes; PPE = **two pairs of chemotherapy-tested gloves (changed every 30-60 min), an impermeable back-closing gown, eye protection and a respirator; never crush or split cytotoxic tablets; pregnant and breast-feeding staff must not handle them; dispose in the yellow cytotoxic stream; keep a spill kit.**
- **Chemical spill:** **evacuate and restrict access → identify the agent from the SDS → don PPE → contain and absorb from the outside inwards → neutralise → bag as hazardous waste → ventilate → report and document.** For a **skin or eye splash**, irrigate at the **eyewash station or shower for a minimum of 15 minutes**.
- **Waste anaesthetic gas control** (the PDF's topic, with the mechanism added): an efficient **scavenger system**, a leak-free circuit with well-fitting masks and connectors, **air-conditioning with HEPA filters that do not recirculate air, anaesthetic gas leak detectors**, and the same ventilation standard extended to the **PACU** because it adjoins the surgical suite.

15. POST-OPERATIVE PATIENT CARE

Complication	Timing and features	Management
Haematoma	24–48 h; swelling, discoloration, pain. A neck haematoma after thyroidectomy is an AIRWAY EMERGENCY	Small → observe; expanding or compressive → open and evacuate (at the bedside for the neck)
Seroma	Day 3–10, especially after mastectomy, axillary or groin dissection, mesh hernia repair ; fluctuant, painless	Aseptic aspiration, compression, drains
Surgical site infection	POD 5–7 ; pain, erythema, warmth, induration, purulent discharge, fever. Spreading cellulitis or crepitus on POD 1–2 → suspect streptococcal or clostridial necrotising infection — a surgical emergency	Open, drain, culture, debride, pack for secondary healing , culture-guided antibiotics — ‘the treatment of pus is drainage’
Dehiscence	POD 5–10 (classically day 7–8) ; the herald sign is a sudden serosanguineous ‘salmon-pink’ discharge with a palpable gap, often after a cough	Abdominal binder, avoid straining, inform the surgeon → usually re-suturing in theatre
Evisceration (burst abdomen)	Protrusion of viscera through the wound	EMERGENCY: stay with the patient; supine, low Fowler's with knees flexed; COVER the viscera with sterile gauze soaked in WARM sterile normal saline — never dry gauze and never push the bowel back; keep nil by mouth; IV access and fluids; monitor for shock; analgesia; notify the surgeon and prepare for immediate theatre
Incisional hernia	Weeks to years later — the late consequence of subclinical dehiscence or infection	Elective mesh repair
Hypertrophic scar vs keloid	Hypertrophic stays within the original margins and tends to regress; keloid extends BEYOND the margins, does not regress and recurs after excision (commoner in darker skin, sternum, shoulder, ear lobe)	Silicone sheeting, pressure, intralesional steroid

15.7 Drains, and the rest of the routine

Drain	Type	Key point
Penrose / corrugated	Passive, open	Gravity or capillary action; higher infection risk
Jackson-Pratt (bulb)	Closed, active	Re-compress the bulb after emptying — a fully expanded bulb means suction has been lost
Hemovac / Redivac	Closed, active (spring / high vacuum)	Orthopaedic, breast and neck surgery

Onset	“W”	Cause	Action
POD 3-5	Water	Urinary tract infection (catheter-associated)	Urinalysis and culture; remove or change the catheter ; antibiotics
POD 4-6	Walking	Deep-vein thrombosis / pulmonary embolism	Calf pain and unilateral swelling, or sudden dyspnoea and hypoxia → Doppler / CTPA; anticoagulate
POD 5-7	Wound	Surgical site infection; also anastomotic leak (with tachycardia and ileus)	Inspect → open, drain and culture ; suspect a leak if the patient is ‘not doing well’
> POD 7	Wonder drugs	Drug fever, abscess/collection, C. difficile colitis, IV-site thrombophlebitis, line infection	Review the drug chart; culture blood, urine and line tips; CT for a collection

MNEMONIC — Wind, Water, Walking, Wound, Wonder drugs
Time order: **Wind (1-2) → Water (3-5) → Walking (4-6) → Wound (5-7) → Wonder drugs (> 7).**
Fever in the first 24 hours is usually NOT infection. Fever after POD 5 with wound signs is infection until proved otherwise.

15.6 Wound healing and wound complications

Phase	Timing	Events	Dominant cell
Haemostasis	Minutes-hours	Vasoconstriction, platelet plug, fibrin clot	Platelets
Inflammatory (Iag)	Day 0-4/5	Vasodilatation, exudate, phagocytosis, debridement; wound held only by clot and sutures	Neutrophils (24-48 h) → MACROPHAGES (48-72 h — the key cell)
Proliferative (fibroplasia)	**Day 4-21	Collagen III deposition, angiogenesis, granulation tissue, epithelialisation, contraction	Fibroblasts
Maturation / remodelling	Day 21 → 1-2 years	Collagen III replaced by collagen I, cross-linking, scar pales and flattens	Fibroblasts, collagenase

- **Tensile strength:** ≈ 5-10 % at 1 week, 30-50 % at 1 month, maximum only 70-80 % of normal skin — which is why dehiscence peaks in the first week and lifting is restricted for six weeks.
- **Types of healing:** **primary (first) intention** — apposed edges, fine scar; **secondary intention** — left open, heals by granulation and contraction (contaminated wounds, abscess cavities); **tertiary / delayed primary closure** — left open 3-5 days then closed.
- **Factors delaying healing** — local: **infection, ischaemia, haematoma, foreign body, tension, dead space, movement, radiation**; systemic: **age, diabetes, malnutrition (protein, vitamin C, vitamin A, zinc), anaemia and hypoxia, obesity, smoking, steroids and immunosuppressants, chemotherapy, jaundice, uraemia, malignancy.**
- **Dressings:** the first dressing over a primarily closed wound is left undisturbed **24-48 h** (epithelialisation is complete and it acts as a bacterial barrier); thereafter change with **aseptic technique**, inspect and document; change immediately if soaked (**strike-through**).
- **Suture removal:** face and neck **3-5 days**; scalp **7-10**; trunk/abdomen **7-10**; upper and lower limb **10-14**; over joints, back, palms and soles **14 days**.

WHY THIS SECTION IS KEPT

Your syllabus heading is “Preparation of Patient for Operation, **Pre & Post Operative Patient Care**”, but Chapter 1 of the PDF ends at the PACU hand-over and only reproduces a blank post-operative form. Everything below is new material.
It has been trimmed to the **examinable core**: PACU and scoring systems, immediate complications, the fever timeline, wound complications and drains. Fluid regimens, nutrition and analgesic pharmacology have been cut as off-syllabus.

15.1 Phases of recovery and the PACU

Phase	Location	Focus
Phase I — immediate	PACU / recovery room, 1 nurse : 1-2 patients	Emergence; airway, breathing, circulation, consciousness, temperature, pain, haemorrhage
Phase II — intermediate	Step-down or ambulatory unit, ward	Preparation for discharge home or to the ward; oral intake, ambulation, teaching
Phase III — extended	Ward, home	Rehabilitation, wound care, complication surveillance

- **Hand-over must include:** procedure and findings, **type of anaesthesia and reversal**, airway status, **allergies**, co-morbidity, intra-operative events and vital-sign trends, **blood loss, fluids and blood products given, urine output**, drains/catheters/packs, **analgesia and antiemetics given with times**, laterality, post-operative orders and whom to call. **The anaesthetist does not leave until the PACU nurse accepts the patient and the hand-over is documented.**
- **Monitoring:** **vital signs every 5-15 min** initially, then 15-minutely until stable — **RR and pattern, SpO₂, BP, pulse, temperature, ECG, pain score, level of consciousness**, plus **hourly urine and drain output**.
- **Position:** **lateral or head-up 15-30°** (keeps the tongue forward, lets secretions drain); **supine flat** after spinal anaesthesia or with hypotension. **Side rails up, oxygen on, suction and an Ambu bag at the bedside, and the patient is never left unattended.**
- **Check:** dressing for ooze, drain patency and character, IV site, **peripheral pulses and capillary refill distal to any cast or tourniquet, ability to move limbs and any paraesthesia** (rules out a positioning injury and assesses spinal regression), and bladder distension.
- **Warm actively** — forced-air blanket, warmed fluids; aim for core temperature > **36 °C** before discharge from PACU.

15.2 Aldrete Score — memorise this table

Parameter	2	1	0
Activity	Moves all four limbs on command	Moves two extremities	Unable to move
Respiration	Breathes deeply and coughs freely	Dyspnoeic or shallow	Apnoeic / ventilated
Circulation (BP)	Within ± 20 % of pre-anaesthetic level	± 20-49 %	± ≥ 50 %
Consciousness	Fully awake	Rousable on calling	Not responding
Oxygen saturation	SpO₂ > 92 % on room air	Needs O ₂ to keep SpO ₂ > 90 %	< 90 % even with O ₂

SCORING — EXAM POINTS
Maximum 10; discharge from PACU requires ≥ 9 with no parameter scoring 0, plus stable vitals, controlled pain and PONV, no active bleeding and normothermia. Mnemonic **A-R-C-C-O**.
The **original 1970 Aldrete score used skin COLOUR**; the **modified score replaced colour with pulse-oximetry SpO₂**.
Pain and nausea are NOT in the Aldrete score — they belong to the **PADSS**. This inversion is a classic MCQ. **PADSS** (Post-Anaesthesia Discharge Scoring System, for discharge **home** after day surgery) scores 0–2 each for **vital signs • ambulation • nausea/vomiting • pain • surgical bleeding; ≥ 9 required**, plus a **responsible escort, written instructions and tolerance of oral fluids**.
Counsel: **no driving, machinery, legal documents, alcohol or sole care of a child for at least 24 hours** after general anaesthesia or sedation.
After spinal/epidural anaesthesia also document **return of motor power, sensation and the sensory level**.

15.3 Immediate post-operative complications

Complication	Cause	Recognition	Management
Airway obstruction — the commonest immediate PACU emergency	Tongue falling back against the posterior pharynx in the sedated supine patient; secretions, blood, laryngeal oedema	Snoring or stridor, see-saw (paradoxical) chest movement, tracheal tug , desaturation	Head tilt-chin lift / jaw thrust , lateral position, suction, oral or nasal airway, 100 % O ₂ ; re-intubate if needed
Laryngospasm	Cord stimulation in light anaesthesia, secretions, blood	Inspiratory stridor progressing to complete silence , paradoxical effort	Remove the stimulus, suction, 100 % O₂ with CPAP and jaw thrust ; if persistent — propofol then succinylcholine and intubate
Hypoventilation	Residual anaesthetic, opioids, incomplete reversal of neuromuscular blockade , pain splinting, obesity, hypothermia	Slow shallow breathing, ↑ EtCO ₂ , drowsiness, pin-point pupils	Stimulate, O ₂ , support ventilation; naloxone 40–80 µg increments for opioids, neostigmine/sugammadex for relaxant, flumazenil for benzodiazepines
Atelectasis — the commonest post-op pulmonary complication	Shallow breathing, retained secretions, upper-abdominal or thoracic incision	Low-grade fever POD 1–2 , reduced basal breath sounds, tachypnoea	Deep breathing and incentive spirometry (10 breaths hourly), splinted coughing, early ambulation, adequate analgesia, physiotherapy — not antibiotics
Hypotension	Hypovolaemia (commonest — blood loss, third-space loss) , residual anaesthetic, sympathetic block from spinal/epidural, arrhythmia, ischaemia, sepsis, anaphylaxis	Check the surgical site and drains for bleeding FIRST ; tachycardia and a narrow pulse pressure precede hypotension	Head-down/legs up, O ₂ , crystalloid bolus 10–20 mL/kg , vasopressor if needed, ECG, Hb, cross-matched blood; surgical control is definitive
Hypertension	Pain (commonest) and bladder distension , hypoxia/hypercarbia, hypothermic shivering, omitted antihypertensives, fluid overload	Raised BP with restlessness	Treat the cause first — analgesia, empty the bladder, oxygenate, warm — before giving antihypertensives

Complication	Cause	Recognition	Management
Delayed emergence	Residual drugs; hypoglycaemia, hypoxia, hypercarbia, hyponatraemia, hypothermia (< 33 °C prevents awakening) ; stroke, seizure	Failure to wake as expected	Check glucose, ABG, electrolytes and pupils ; reverse opioids and benzodiazepines; think stroke if there are focal signs
Emergence delirium / agitation	Commonest in children and the elderly ; also hypoxia, pain, full bladder, ketamine, withdrawal	Restlessness, disorientation, thrashing	Always exclude hypoxia first , then pain and a full bladder; reassure and orientate; avoid physical restraint
Hypothermia and shivering	Cold OR, anaesthetic vasodilatation, cold fluids, large exposed cavity, elderly and neonates	Core < 36 °C; shivering raises O₂ consumption up to 400 % ; causes coagulopathy, ↑ blood loss, ↑ SSI, arrhythmia, delayed drug metabolism	Forced-air warming, warmed fluids and irrigation, raised room temperature ; oxygen; pethidine (meperidine) 12.5–25 mg IV is the most effective drug for shivering
Haemorrhage	Reactionary — within 24 h (slipped ligature, clot dislodged as BP rises); Secondary — POD 7–14 from infection eroding a vessel	Tachycardia, narrow pulse pressure, restlessness, cool clammy skin, oliguria, brisk drain output	Two large-bore cannulae, crystalloid, cross-matched blood / massive transfusion protocol , tranexamic acid, keep warm, return to theatre

15.4 Post-operative nausea and vomiting (PONV) — in brief

- Incidence **20–30 %**, up to **80 %** in high-risk patients; the commonest reason for unplanned admission after day surgery.
- Apfel score — 1 point each for: female sex • non-smoker • previous PONV or motion sickness • expected post-operative opioid use**. Risk for 0/1/2/3/4 factors ≈ **10 / 20 / 40 / 60 / 80 %**. **Smoking is protective**.
- Other risks: volatile agents and **nitrous oxide**, long anaesthesia, and gynaecological, laparoscopic, middle-ear, squint and breast surgery.
- Drugs by receptor: 5-HT₃ antagonist — ondansetron 4–8 mg IV, given at the END of surgery (watch QT prolongation) • steroid — dexamethasone 4–8 mg at induction • D₂ antagonist — metoclopramide** (extrapyramidal effects; avoid in obstruction) • **NK₁ — aprepitant • antihistamine — cyclizine • anticholinergic — hyoscine patch • propofol TIVA** is itself antiemetic.
- Strategy: 1–2 risk factors → two agents from DIFFERENT classes; ≥ 3 → three or four agents plus propofol TIVA, avoiding nitrous oxide and sparing opioids**. For breakthrough PONV use a class **not already given**.

15.5 Post-operative fever — the “W’s” timeline (very high yield)

Onset	“W”	Cause	Action
During or within 24 h	What we did / Wonder drugs	Inflammatory (cytokine) response to surgery, drug or transfusion reaction, MALIGNANT HYPERTHERMIA , pre-existing infection	Usually NOT infective . Rigidity + ↑ EtCO ₂ → dantrolene; stop any transfusion
POD 1–2	Wind	ATELECTASIS — the commonest cause of early post-op fever ; aspiration, early pneumonia	Incentive spirometry, deep breathing, ambulation, analgesia, physiotherapy

Q17. In the lateral position the axillary (chest) roll should be placed

(a) Directly in the axilla (b) Just caudal to the axilla under the chest wall (c) Under the iliac crest (d) Between the knees

Ans: (b) Just caudal to the axilla under the chest wall — It goes **caudal to (below) the axilla** to lift the chest wall off the **dependent brachial plexus and axillary vessels**. Placing it **in** the axilla causes the very injury it should prevent.

C — Environment, Asepsis and SSI

Q18. A HEPA filter removes

(a) 99.97 % of particles $\geq 0.3 \mu\text{m}$ (b) 99.9 % of particles $\geq 1 \mu\text{m}$ (c) 95 % of particles $\geq 0.5 \mu\text{m}$ (d) 99.999 % of particles $\geq 0.1 \mu\text{m}$

Ans: (a) 99.97 % of particles $\geq 0.3 \mu\text{m}$ — HEPA = 99.97 % of particles $0.3 \mu\text{m}$ and larger. The 99.999 % at $0.1 \mu\text{m}$ figure is an **ULPA** filter, used in surgical-smoke evacuators.

Q19. The operating room is maintained at which pressure relative to the corridor?

(a) Negative (b) Positive (c) Equal (d) Alternating

Ans: (b) Positive — Positive, so air flows **out** of the OR. **Negative pressure with ≥ 12 air changes/hour** is used for **airborne isolation — tuberculosis, measles, varicella**.

Q20. The recommended relative humidity in an operating room is

(a) Below 20 % (b) 20–60 % (c) 70–80 % (d) Above 80 %

Ans: (b) 20–60 % — 20–60 % (classically 30–60 %). **Too low $\rightarrow$ static electricity and spark risk; too high $\rightarrow$ microbial growth and condensation.** Temperature **20–23 °C** with **≥ 15 air changes/hour**.

Q21. The single most effective measure for preventing healthcare-associated infection is

(a) Prophylactic antibiotics (b) Hand hygiene (c) Ultraviolet irradiation (d) Routine fumigation

Ans: (b) Hand hygiene — Hand hygiene with an antimicrobial agent — stated explicitly in your PDF as the greatest single factor in preventing nosocomial infection. Routine formaldehyde fumigation is obsolete and UV is only an adjunct.

Q22. A draped instrument table is considered sterile

(a) To 30 cm below the table top (b) Only at and above table-top level (c) Down to the floor (d) Only in the central third

Ans: (b) Only at and above table-top level — Anything below table-top level is unsterile and is never brought back up. Related: the wrapper edge (**1 inch / 2.5 cm**) is unsterile, the **gown back** is unsterile, and **moisture contaminates by strike-through**.

Q23. An elective cholecystectomy without spillage is classified as

(a) Clean (b) Clean-contaminated (c) Contaminated (d) Dirty

Ans: (b) Clean-contaminated — Entry into the biliary/GI tract under controlled conditions = **clean-contaminated (Class II), SSI 3–10 %**. Clean < **2 %**, contaminated **15–20 %**, dirty/infected **30–40 %**.

Q24. The commonest organism causing surgical site infection is

(a) Escherichia coli (b) Pseudomonas aeruginosa (c) Staphylococcus aureus (d) Clostridium perfringens

Ans: (c) Staphylococcus aureus — Staphylococcus aureus, then coagulase-negative staphylococci, Enterococcus and E. coli. The commonest **source** of contamination is the **patient's own endogenous flora**.

Q25. Alcohol-based hand rub is ineffective, and soap and water must be used, against

(a) MRSA (b) Clostridioides difficile (c) Influenza virus (d) Pseudomonas aeruginosa

Ans: (b) Clostridioides difficile — Alcohol is not sporicidal, so **C. difficile spores need mechanical removal with soap and water** (plus contact precautions and hypochlorite for surfaces). Soap and water is also mandatory for **visibly soiled hands**.

Q26. Artificial fingernails are prohibited in the OR chiefly because they

- (a) Tear surgical gloves (b) Harbour Gram-negative bacteria and fungi (c) Interfere with pulse oximetry (d) Absorb chlorhexidine

Ans: (b) Harbour Gram-negative bacteria and fungi — AORN prohibits artificial nails and chipped polish because the **subungual space harbours Gram-negative bacilli and fungi** and cannot be decontaminated. (Polish does also affect pulse oximetry, but infection control is the reason asked.)

Q27. The correct sequence for REMOVING (doffing) personal protective equipment is

- (a) Gown → gloves → mask → goggles (b) Gloves → goggles → gown → mask (c) Mask → gown → gloves → goggles (d) Goggles → gloves → mask → gown

Ans: (b) Gloves → goggles → gown → mask — **Doffing: gloves → goggles/face shield → gown → mask/respirator (outside the room) → hand hygiene.** The most contaminated item comes off first and the respirator last. **Donning is the reverse: gown → mask → goggles → gloves.**

D — Electro Surgery and Equipment

Q28. Electrosurgical units operate in which frequency range?

- (a) 50–60 Hz (b) 1–10 kHz (c) 300 kHz – 3 MHz (d) 10–100 MHz

Ans: (c) 300 kHz - 3 MHz — Radiofrequency, 300 kHz - 3 MHz. Above about **100 kHz** the current no longer stimulates nerve or muscle (**the Faradic effect is avoided**), which is why RF current cuts and coagulates without electrocuting.

Q29. Which statement about bipolar electrosurgery is TRUE?

- (a) It requires a dispersive return pad (b) Current passes through the whole patient (c) No dispersive pad is required (d) It cannot be used in patients with pacemakers

Ans: (c) No dispersive pad is required — Both electrodes are the **two tines of the forceps**, so current passes only between them. **No pad is needed**, lateral spread is minimal, and it is the **safest mode with a pacemaker**.

Q30. The coagulation waveform of an ESU is

- (a) Continuous low-voltage current (b) Interrupted high-voltage current (c) Direct current (d) Continuous current at 50 Hz

Ans: (b) Interrupted high-voltage current — **Coag = interrupted, high-voltage, low duty cycle (~6 %); cut = continuous, low-voltage, high-current (100 % duty cycle).** Coag's high voltage explains its greater risk of **capacitive coupling and insulation breakdown**.

Q31. Capacitive coupling in laparoscopic electrosurgery is most likely when

- (a) Bipolar forceps are used (b) A metal instrument passes through a plastic (hybrid) trocar (c) The pad is on the thigh (d) Cut mode is used at low power

Ans: (b) A metal instrument passes through a plastic (hybrid) trocar — **A hybrid metal-plastic trocar** lets current be induced through **intact insulation** into surrounding tissue. Use all-metal or all-plastic systems, lowest power, cut rather than coag, short activations and **active-electrode monitoring**.

Q32. A pin-hole break in the insulation of a laparoscopic electrode classically causes

- (a) An immediately visible spark (b) A return-electrode burn (c) Delayed bowel perforation several days later (d) Failure of the generator to activate

Ans: (c) Delayed bowel perforation several days later — **Insulation failure** discharges current **outside the surgeon's field of view** → a full-thickness bowel burn perforating **3–10 days post-operatively** with peritonitis. Inspect insulation before every use.

15. HEPA = 99.97 % at 0.3 µm; the OR is **positive** pressure with **≥ 15 air changes/hour**.
16. Hand hygiene is the **single most effective measure** against healthcare-associated infection.
17. A **draped table is sterile only at table-top level**; the wrapper edge (**1 inch**) is unsterile; **moisture = contamination**.
18. **X-ray-detectable sponges are never used as dressings; nothing leaves the OR until the final count is correct.**
19. **MH: volatile agents + succinylcholine; dantrolene 2.5 mg/kg; earliest sign is rising EtCO₂.**
20. **Anaphylaxis: adrenaline 0.5 mg IM 1:1000; latex reactions are delayed 20-60 min.**
21. The commonest **ignition** source of OR fires is the **ESU**; the commonest **fuel** is **alcohol prep**; **CO₂ for electrical fires**.
22. **Aldrete ≥ 9** to leave PACU; **PADSS ≥ 9** to go home; **pain is in PADSS, not Aldrete**.
23. Commonest immediate PACU emergency = **airway obstruction from the tongue**; commonest post-op pulmonary complication = **atelectasis**.
24. **Dehiscence POD 5-10 with serosanguineous discharge; evisceration → warm sterile saline-soaked gauze and call the surgeon.**
25. **Radiation: 20 mSv/year; doubling the distance quarters exposure; the dosimeter goes OUTSIDE the apron.**

End of exam-focused notes • Revise the PDF for the Chapter 1-3 core text, and these 18 sections for every gap.

Q33. A return-electrode contact quality monitoring (REM) system prevents

- (a) Alternate-site burns from metal contact (b) Pad-site burns from inadequate pad contact (c) Capacitive coupling (d) Surgical smoke exposure

Ans: (b) Pad-site burns from inadequate pad contact — A split pad whose halves are monitored for impedance; the generator **shuts down when contact becomes inadequate**, virtually eliminating **return-electrode (pad-site) burns**.

Q34. The surgeon repeatedly asks for higher ESU power. The correct action is to

- (a) Increase the power as requested (b) Switch to bipolar (c) Stop, check all connections and the dispersive pad, and replace the unit if faulty (d) Reposition the ECG electrodes

Ans: (c) Stop, check all connections and the dispersive pad, and replace the unit if faulty — Repeated requests indicate a **fault — poor pad contact, a loose connection, a frayed cord or a failing generator**. Check the circuit; if no cause is found, **turn off and unplug the unit, label it, get another, and document the occurrence and serial numbers**.

Q35. Which statement about surgical smoke is correct?

- (a) A standard surgical mask is adequate protection (b) Ablating 1 g of tissue is roughly equivalent to 3-6 unfiltered cigarettes (c) It is sterile and harmless (d) A HEPA rather than an ULPA filter is required in the evacuator

Ans: (b) Ablating 1 g of tissue is roughly equivalent to 3-6 unfiltered cigarettes — Smoke contains **> 150 chemicals** plus viable cells and viral DNA (**documented HPV transmission to surgeons**). Control = **smoke evacuator with an ULPA filter held within 1-2 inches** plus a **high-filtration mask or N95 —** a standard mask is **not** sufficient.

Q36. For an average adult, maximum recommended tourniquet time and pressure for the LOWER limb are

- (a) 30 min at 200 mmHg (b) 60 min at 250 mmHg (c) 90-120 min at 300-350 mmHg (d) 3 hours at 400 mmHg

Ans: (c) 90-120 min at 300-350 mmHg — Lower limb 300-350 mmHg for ≤ 90-120 min; upper limb 250-300 mmHg for ≤ 60 min. Deflate for 10-15 min if longer, and record site, pressure and both inflation and deflation times.

E — Sterilization and Disinfection

Q37. Standard gravity-displacement autoclave conditions are

- (a) 100 °C at atmospheric pressure for 30 min (b) 121 °C at 15 psi for 15-20 min (c) 134 °C at 15 psi for 30 min (d) 160 °C for 2 hours

Ans: (b) 121 °C at 15 psi for 15-20 min — 121 °C (250 °F) at 15 psi for 15-20 minutes, or 134 °C at 30 psi for 3-3.5 minutes. 160 °C for 2 h is a **hot air oven (dry heat)** cycle.

Q38. Flash (immediate-use) steam sterilization of an unwrapped metal instrument is carried out at

- (a) 121 °C for 20 min (b) 132 °C for 3 min (c) 132 °C for 30 min (d) 160 °C for 1 hour

Ans: (b) 132 °C for 3 min — 132 °C (270 °F) for 3 minutes unwrapped non-porous metal; **10 minutes** for porous, lumened or complex items. IUSS is **only for an urgently needed single item —** never implants, sets, or convenience.

Q39. The Bowie-Dick test detects

- (a) Inadequate air removal in a prevacuum sterilizer (b) Adequate ethylene oxide concentration (c) Spore-killing efficacy (d) Steam superheating only

Ans: (a) Inadequate air removal in a prevacuum sterilizer — A Class 2 chemical indicator run **daily in an EMPTY prevacuum chamber at 134 °C for 3.5 min** to test **air removal / steam penetration**. It does **not** prove sterility — only a **biological indicator** does.

Q40. The biological indicator for steam sterilization is

(a) Bacillus atrophaeus (b) Geobacillus stearothermophilus (c) Bacillus pumilus (d) Clostridium sporogenes
Ans: (b) Geobacillus stearothermophilus — Geobacillus stearothermophilus (thermophile, incubated 55–60 °C) for **steam and hydrogen peroxide plasma**; **Bacillus atrophaeus** for **dry heat and ethylene oxide**; **Bacillus pumilus** for **gamma radiation**.

Q41. Dry heat sterilization in a hot air oven requires

(a) 121 °C for 15 min (b) 134 °C for 3 min (c) 160 °C for 2 hours (d) 180 °C for 2 hours
Ans: (c) 160 °C for 2 hours — 160 °C for 2 h (or 170 °C for 1 h, 180 °C for 30 min). It is preferred for **glassware, oils, powders, greases and sharp instruments — it does not blunt or corrode them —** but not for rubber, plastics or linen.

Q42. Which item CANNOT be processed in a hydrogen peroxide gas plasma (STERRAD) sterilizer?

(a) A fibre-optic cable (b) A stainless-steel instrument (c) Cotton gauze and paper drapes (d) A microsurgical instrument
Ans: (c) Cotton gauze and paper drapes — Cellulose — paper, cotton, linen, gauze, dressings — absorbs the sterilant, as do liquids, powders and long narrow lumens. Advantages: **45-50 °C, ≈ 1 h cycle, and NO aeration** because by-products are only water and oxygen.

Q43. Sterilization with 2 % activated glutaraldehyde requires immersion for

(a) 10 minutes (b) 30 minutes (c) 3 hours (d) 10 hours
Ans: (d) 10 hours — 10 hours for sterilization; high-level disinfection in 20-45 minutes (your PDF quotes 10 min — quote the textbook if the question comes from it). Rinse in **sterile distilled water** and **test potency with a strip** before each use.

Q44. After use, peracetic acid decomposes into

(a) Formaldehyde and water (b) Acetic acid, water and oxygen (c) Chlorine and water (d) Ethylene glycol
Ans: (b) Acetic acid, water and oxygen — Acetic acid (vinegar), water and oxygen — so it is environmentally safe and can go down an ordinary drain. It also works **in the presence of organic soil** and its low surface tension reaches lumens.

Q45. Under the Spaulding classification, a flexible gastrointestinal endoscope is

(a) Critical — needs sterilization (b) Semi-critical — needs high-level disinfection (c) Non-critical — needs low-level disinfection (d) Non-critical — needs only cleaning
Ans: (b) Semi-critical — needs high-level disinfection — Endoscopes contact **intact mucous membranes → semi-critical → high-level disinfection. Critical** items enter sterile tissue or the vascular system (sterilization); **non-critical** items touch only intact skin.

Q46. The internationally accepted Sterility Assurance Level is

(a) 10⁻³ (b) 10⁻⁴ (c) 10⁻⁶ (d) 10⁻¹²
Ans: (c) 10⁻⁶ — SAL = 10⁻⁶ — no more than one viable organism per million sterilized items. This is the quantitative definition of ‘sterile’.

Q47. Which is the MOST resistant to sterilization and disinfection?

(a) HIV (b) Mycobacterium tuberculosis (c) Bacterial spores (d) Prions
Ans: (d) Prions — Decreasing resistance: **prions > bacterial spores > coccidia > mycobacteria > small non-lipid viruses > fungi > vegetative bacteria > lipid-enveloped viruses (HIV, HBV — the easiest to kill)**. Prions need **1 N NaOH plus 134 °C for 18 min**.

Q48. The shelf-life of a sterilized wrapped pack is determined by

(a) A fixed 7-day limit (b) A fixed 30-day limit (c) Event-related sterility (d) The class of chemical indicator used
Ans: (c) Event-related sterility — Event-related sterility — the pack is sterile until an event compromises it (wet, torn, punctured, seal broken, dropped, over-handled). The old “30 days” rule is obsolete.

Mnemonic	Stands for
Geo = Steam ; Atro = Air & EO	Geobacillus stearothermophilus → steam and plasma; Bacillus atrophaeus → dry heat (air) and Ethylene oxide
R - A - C - E	Fire: R escue, A larm, C onfine, E xtinguish/Evacuate
P - A - S - S	Extinguisher: P ull, A im at the base, S queeze, S weep
R - P - M	START triage: R espiration, P erfusion, M ental status
4 H's & 4 T's	Reversible causes of arrest: Hypoxia, Hypovolaemia, Hypo/Hyperkalaemia-metabolic, Hypothermia; Tension pneumothorax, Tamponade, Toxins, Thrombosis
4 D's	Negligence: D uty, D ereliction, D irect causation, D amage
A - R - C - C - O	Aldrete: A ctivity, R espiration, C irculation, C onsciousness, O xygen saturation — max 10, discharge ≥ 9
Wind - Water - Walking - Wound - Wonder drugs	Post-op fever by day: 1-2 → 3-5 → 4-6 → 5-7 → > 7
Time - Distance - Shielding	Radiation protection, under ALARA
Yellow burns • Red recycles • White is sharp • Blue is glass	Biomedical waste colours
‘Only what is documented is a legal fact’	Perioperative documentation
‘When in doubt, throw it out’	Doubtful sterility = unsterile
‘The treatment of pus is drainage’	Surgical site infection

The 25 statements most likely to be a one-mark question

1. The **circulator** keeps the legal record of the count and is the **patient's advocate**.
2. **At least two identifiers**, and the patient **states** their own name.
3. The site is marked by the **operating surgeon** with an **indelible marker**, with the patient participating.
4. The **time out** is **after induction, before skin incision**; the WHO checklist has **3 phases**.
5. **Hearing is the last sense to be lost —** silence during induction.
6. The **safety strap across the mid-thighs** is the key measure against a fall from the table.
7. **Ulnar nerve —** commonest peri-operative nerve injury; **common peroneal** in lithotomy (**foot drop**).
8. **Trendelenburg raises ICP and IOP and lowers FRC; Kraske/jackknife is a PRONE modification**.
9. **Never move an anaesthetised patient without the anaesthetist's permission**.
10. The patient's **skin must never touch metal** when the ESU is in use (alternate-site burn), and the **pencil stays in its holster**.
11. ESU works at **300 kHz-3 MHz; bipolar needs no pad** and is safest with a pacemaker; **coag = interrupted high voltage**.
12. Autoclave 121 °C / 15 psi / 15-20 min; hot air oven 160 °C / 2 h; flash 132 °C / 3 min.
13. **Bowie-Dick tests air removal**; the **biological indicator is the only proof of sterilizing efficacy; SAL = 10⁻⁶**.
14. **EO requires aeration; plasma cannot be used for cellulose; glutaraldehyde 10 h sterilizes**.

Q72. The correct immediate action for post-operative wound evisceration is to
(a) Push the viscera back and apply a dry dressing (b) Cover the viscera with sterile gauze soaked in warm sterile saline, position low Fowler's with knees flexed, keep nil by mouth and call the surgeon (c) Sit the patient upright and give oral fluids (d) Apply a tight binder and discharge
Ans: (b) Cover the viscera with sterile gauze soaked in warm sterile saline, position low Fowler's with knees flexed, keep nil by mouth and call the surgeon — Never replace the viscera and never use dry gauze. Cover with **warm sterile saline-moistened gauze**, position supine/low Fowler's with knees flexed to reduce tension, keep **nil by mouth**, secure IV access, monitor for shock and **prepare for immediate theatre**.

Q73. Sutures on the face are typically removed on day
(a) 3-5 (b) 7-10 (c) 10-14 (d) 14-21
Ans: (a) 3-5 — Face and neck 3-5 days (rich blood supply, cosmesis); scalp and trunk **7-10**; limbs **10-14**; joints, back, palms and soles **14 days**.

Q74. Post-operative return of bowel function occurs in which order?
(a) Colon → stomach → small bowel (b) Small bowel → stomach → colon (c) Stomach → colon → small bowel (d) All simultaneously at 72 hours
Ans: (b) Small bowel → stomach → colon — Small intestine ≈ 24 h → stomach 24-48 h → COLON 48-72 h (last to recover). Resolution is judged by the **passage of flatus or stool**, not bowel sounds.

Q75. Which is a CLOSED active suction drain?
(a) Penrose drain (b) Corrugated drain (c) Jackson-Pratt drain (d) Open sump drain
Ans: (c) Jackson-Pratt drain — Jackson-Pratt (bulb) and Hemovac/Redivac (spring or high vacuum) are closed active systems with a lower infection risk; **Penrose and corrugated drains are passive and open**. A fully re-expanded JP bulb means suction has been lost — re-compress it after emptying.

Q76. According to the Apfel score, which is NOT a risk factor for PONV?
(a) Female sex (b) Non-smoking status (c) Smoking (d) Expected post-operative opioid use
Ans: (c) Smoking — The four Apfel factors are **female sex, non-smoker, previous PONV or motion sickness, and expected post-operative opioid use** — giving ≈ **10/20/40/60/80 %** risk for 0-4 factors. **Smoking is protective**.

Q77. Secondary haemorrhage after surgery classically occurs on which day, and why?
(a) Day 0; slipped ligature (b) Within 24 h; clot dislodged as blood pressure rises (c) POD 7-14; infection eroding a vessel wall (d) POD 30; false aneurysm
Ans: (c) POD 7-14; infection eroding a vessel wall — Primary = during surgery; **reactionary** = within 24 h (slipped ligature or clot dislodged as BP rises and vasoconstriction wears off); **secondary** = **POD 7-14 from infection eroding the vessel wall**.

18. RAPID REVISION

Mnemonics

Mnemonic	Stands for
2 - 4 - 6 - 8	Fasting: clear fluids 2 h, breast milk 4 h, formula/light meal 6 h, fatty meal 8 h
Sign In - Time Out - Sign Out	The 3 phases of the WHO Surgical Safety Checklist
PADDING	Positioning: P ressure points padded, A lignment, D ignity, D evelopes ready, I V lines protected, N erves free, G entle movement
Critical - Semi-critical - Non-critical	Spaulding: sterilize - high-level disinfect - low-level disinfect

Q49. Gamma radiation sterilization uses a cobalt-60 source at a dose of
(a) 2.5 kGy (b) 25 kGy (c) 250 kGy (d) 2500 kGy
Ans: (b) 25 kGy — 25 kGy (2.5 Mrad) — a cold, industrial method for pre-packed **single-use disposables** (syringes, gloves, catheters, sutures, blades, heart valves), **not used inside hospitals** because of cost and shielding.

Q50. Membrane filtration of a heat-labile solution uses a pore size of
(a) 0.45 µm (b) 0.22 µm (c) 1.2 µm (d) 5 µm
Ans: (b) 0.22 µm — 0.22 µm retains bacteria and is used for sera, antibiotic and vaccine solutions and media. Note it **removes but does not kill**, and **does not retain viruses or mycoplasma** (0.1 µm is needed).

F — Emergencies and Disasters

Q51. Malignant hyperthermia is triggered by
(a) Propofol and nitrous oxide (b) Volatile inhalational agents and succinylcholine (c) Non-depolarising muscle relaxants (d) Local anaesthetics
Ans: (b) Volatile inhalational agents and succinylcholine — Triggers are **all volatile agents and succinylcholine**. **Safe: propofol, thiopentone, benzodiazepines, opioids, nitrous oxide, non-depolarising relaxants and all local anaesthetics**. The defect is in the **RYR1 gene on chromosome 19**.

Q52. The initial dose of dantrolene in malignant hyperthermia is
(a) 0.5 mg/kg (b) 1 mg/kg (c) 2.5 mg/kg (d) 10 mg/kg immediately
Ans: (c) 2.5 mg/kg — 2.5 mg/kg IV bolus, repeated every 5-10 min up to 10 mg/kg (occasionally 30). Simultaneously stop the triggers, **hyperventilate with 100 % O₂ at ≥ 10 L/min**, cool actively (**stop at 38 °C**), treat hyperkalaemia, and keep urine output > **1-2 mL/kg/h**.

Q53. The earliest sign of malignant hyperthermia under general anaesthesia is
(a) Hyperthermia (b) An unexplained rise in end-tidal CO₂ (c) Myoglobinuria (d) Hypotension
Ans: (b) An unexplained rise in end-tidal CO₂ — Rising EtCO₂ with tachycardia and tachypnoea; **masseter spasm** after succinylcholine is the classic clue. **Hyperthermia is a LATE sign** — waiting for it costs lives.

Q54. The adult intramuscular dose of adrenaline in anaphylaxis is
(a) 0.1 mg of 1:10 000 (b) 0.5 mg of 1:1000 into the antero-lateral thigh (c) 1 mg of 1:1000 subcutaneously (d) 5 mg nebulised
Ans: (b) 0.5 mg of 1:1000 into the antero-lateral thigh — 0.5 mg IM (0.5 mL of 1:1000) into the antero-lateral thigh, repeated every 5 min; 50 µg IV boluses if monitored and experienced. Then **crystalloid 20 mL/kg**, chlorpheniramine, hydrocortisone and salbutamol.

Q55. Latex anaphylaxis under anaesthesia characteristically
(a) Never causes hypotension (b) Is typically delayed 20-60 minutes after exposure (c) Occurs only in adults (d) Responds only to steroids
Ans: (b) Is typically delayed 20-60 minutes after exposure — Absorption across **mucosa or peritoneum** makes latex reactions **delayed 20-60 min** rather than immediate. Highest risk: **spina bifida, children with multiple operations, health-care workers**. Cross-reacting foods: **banana, avocado, chestnut, kiwi**.

Q56. The extinguisher of choice for an electrical fire in the OR is
(a) Water (b) Foam (c) Carbon dioxide (Class C) (d) Class D dry powder
Ans: (c) Carbon dioxide (Class C) — CO₂ or ABC dry chemical for Class C electrical fires — never water. CO₂ is preferred around equipment because it leaves no residue. The commonest **ignition** source of OR fires is the **ESU (70-90 %)** and the commonest **fuel** is **alcohol-based prep**.

Q57. In START triage, a casualty who is not breathing even after the airway is opened is tagged

- (a) Red (b) Yellow (c) Green (d) Black

Ans: (d) Black — Black = dead/expectant. Red = immediate (RR > 30, absent radial pulse or capillary refill > 2 s, or not obeying commands); **Yellow** = delayed; **Green** = minor/walking. START assesses **RPM — Respiration, Perfusion, Mental status** in under a minute.

G — Radiation, Chemical Hazards and Waste

Q58. The SI unit of equivalent dose of radiation is the

- (a) Gray (b) Sievert (c) Becquerel (d) Roentgen

Ans: (b) Sievert — Sievert (Sv) — used for **all dose limits and personnel monitoring (1 Sv = 100 rem)**. The **Gray** is absorbed dose (1 Gy = 100 rad) and the **Becquerel** is activity.

Q59. The annual occupational whole-body radiation dose limit is

- (a) 1 mSv (b) 20 mSv averaged over 5 years (c) 50 mSv every year without restriction (d) 150 mSv

Ans: (b) 20 mSv averaged over 5 years — 20 mSv/year averaged over 5 consecutive years (100 mSv/5 y), with a maximum of 50 mSv in any single year. Public 1 mSv/y; lens 20 mSv/y; extremities 500 mSv/y; declared pregnancy ≤ 1 mSv to the fetus (0.5 mSv/month).

Q60. If the distance from a radiation source is doubled, exposure is reduced to

- (a) One half (b) One quarter (c) One eighth (d) One sixteenth

Ans: (b) One quarter — Inverse square law — intensity ∝ 1/d², so doubling gives one quarter and tripling gives one ninth. Stand ≥ 2 m (6 ft) away or leave the room. The three principles are time, distance and shielding, under ALARA.

Q61. A personal radiation dosimeter should be worn

- (a) Inside the lead apron at waist level (b) Outside the lead apron at chest/collar level (c) On the wrist (d) On the X-ray machine

Ans: (b) Outside the lead apron at chest/collar level — Outside the apron at chest or collar level, to record dose to the unshielded neck, head and lens. A **pregnant** worker wears a **second badge under the apron at waist level**. **TLD** is the commonest type.

Q62. Radiation-induced carcinogenesis is an example of a

- (a) Deterministic effect with a threshold (b) Stochastic effect with no threshold (c) Acute deterministic effect (d) Non-radiation effect

Ans: (b) Stochastic effect with no threshold — Stochastic effects (cancer, heritable mutation) have no threshold — probability rises with dose. Deterministic effects (erythema, epilation, cataract, sterility, marrow suppression) have a threshold and their severity rises with dose.

Q63. The OSHA permissible exposure limit (8-h TWA) for ethylene oxide is

- (a) 0.05 ppm (b) 1 ppm (c) 25 ppm (d) 100 ppm

Ans: (b) 1 ppm — EO = 1 ppm (excursion 5 ppm/15 min). Others: **glutaraldehyde ceiling 0.05 ppm, formaldehyde 0.75 ppm, nitrous oxide 25 ppm (NIOSH), halogenated volatiles 2 ppm, methyl methacrylate 100 ppm.**

Q64. Under the Biomedical Waste Management Rules 2016, a used syringe with a fixed needle is discarded into a

- (a) Yellow bag (b) Red bag (c) White translucent puncture-proof container (d) Blue container

Ans: (c) White translucent puncture-proof container — Sharps, including syringes with fixed needles, go in the WHITE translucent puncture-proof container → autoclaved and shredded, then encapsulated or sent to a foundry. **Red** = recyclable plastics (syringes **without** needles); **Yellow** = anatomical, soiled, expired drugs, cytotoxics; **Blue** = glass and metallic implants.

H — Post-Operative Patient Care

Q65. Which of the following is NOT a component of the modified Aldrete score?

- (a) Respiration (b) Circulation (c) Pain (d) Oxygen saturation

Ans: (c) Pain — The modified Aldrete score assesses **Activity, Respiration, Circulation, Consciousness and Oxygen saturation (A-R-C-C-O)**, each 0-2, maximum 10, **discharge ≥ 9. Pain and nausea belong to the PADSS**, used for discharge home after day surgery.

Q66. The original Aldrete score used which parameter that the modified score replaced with pulse oximetry?

- (a) Temperature (b) Skin colour (c) Urine output (d) Bowel sounds

Ans: (b) Skin colour — The 1970 score used **skin colour**; the modified score substituted **SpO₂**, which is objective and more sensitive.

Q67. The commonest cause of upper airway obstruction in the immediate post-anaesthesia period is

- (a) Laryngospasm (b) The tongue falling back against the posterior pharynx (c) Bronchospasm (d) Vocal cord paralysis

Ans: (b) The tongue falling back against the posterior pharynx — Loss of pharyngeal tone in the sedated supine patient lets the **tongue fall backwards** — recognised by **snoring and see-saw (paradoxical) chest movement**. Treat with **jaw thrust or head tilt-chin lift, lateral position, suction and an airway**.

Q68. The commonest post-operative pulmonary complication is

- (a) Pneumonia (b) Atelectasis (c) Pulmonary embolism (d) Pneumothorax

Ans: (b) Atelectasis — Atelectasis — typically low-grade fever on **POD 1-2** with reduced basal breath sounds. Treatment is **deep breathing, incentive spirometry, splinted coughing, early ambulation and adequate analgesia** — not antibiotics.

Q69. Fever on the second post-operative day is most likely due to

- (a) Wound infection (b) Atelectasis (c) Deep vein thrombosis (d) Urinary tract infection

Ans: (b) Atelectasis — Follow the W's in time order: **Wind (atelectasis, POD 1-2) → Water (UTI, POD 3-5) → Walking (DVT, POD 4-6) → Wound (SSI, POD 5-7) → Wonder drugs (> POD 7)**. Fever in the **first 24 h is usually not infective**.

Q70. Fever on the sixth post-operative day with wound erythema and purulent discharge indicates

- (a) Atelectasis (b) Drug fever (c) Surgical site infection (d) Malignant hyperthermia

Ans: (c) Surgical site infection — SSI classically declares itself on POD 5-7. Management: open, drain and culture the wound, debride and pack for healing by secondary intention, with culture-guided antibiotics.

Q71. Wound dehiscence most commonly occurs on which post-operative day, and its herald sign is

- (a) POD 1-2; bright red bleeding (b) POD 5-10; serosanguineous 'salmon-pink' discharge (c) POD 14-21; purulent discharge (d) POD 30; a palpable lump

Ans: (b) POD 5-10; serosanguineous 'salmon-pink' discharge — Dehiscence peaks at **POD 5-10 (classically day 7-8)**, when the wound has only **5-10 %** of its final tensile strength. The warning sign is a **sudden serosanguineous (salmon-pink) discharge** with a palpable gap.